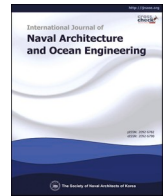




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Method for collision avoidance based on deep reinforcement learning with path-speed control for an autonomous ship

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ABSTRACT

In this paper, we propose a collision avoidance method based on deep reinforcement learning (DRL) that simultaneously controls the path and speed of a ship. The DRL is actively applied in machine control and artificial intelligence. To verify the proposed method, we applied it to the Imazu problem. It provides benchmark scenarios for collision avoidance. In particular, we compared and analyzed the collision avoidance performance according to the level of learning and various parameters to ensure that the proposed method displays optimal avoidance performance. The results indicated that the proposed method can determine a safe avoidance path for a given situation. Finally, to compare the performance of the proposed method, we compared the collision avoidance method based on the path-speed control of the OS proposed in this study with the collision avoidance method that controls only the path of the OS (Chun et al., 2021). We observed that the proposed method failed in 6 out of 20 scenarios of the Imazu problem when only the path of the OS was controlled. However, it succeeded in collision avoidance in all the 20 scenarios when both path and speed were controlled simultaneously.

1. Introduction

1.1. Research background

As shipping routes become increasingly complex, the number of human-caused ship accidents (such as by navigator inattention) is also increasing. In particular, in coastal areas where many ships operate in small regions, there is a growing need for an autonomous navigation system that accurately predicts the potential risks of own ship (OS) and performs safe navigation without collision with target ships (TSs). To achieve this, it is necessary to accurately identify the position, speed, and direction of maritime obstacles such as TSs in real-time therefore, assess the collision risk of each TS and perform collision avoidance at an appropriate time.

All ships are required to perform collision avoidance in compliance with the Convention on International Regulations for Preventing Collisions at Sea (COLREGs, 1972) to safely prevent collisions between ships during navigation. Many studies, including our previous work, have proposed collision avoidance methods that apply deep reinforcement learning (DRL) to autonomous navigation systems that comply with the

COLREGs and can flexibly respond to abrupt variations in conditions (Chun et al., 2021; He et al., 2017; Zhang et al., 2015; Zhao et al., 2016). However, the collision avoidance method applied to the existing autonomous navigation system could not achieve collision avoidance in a complex environment by performing collision avoidance based only on the most hazardous TS. It was limited to controlling the path (and not the speed) by considering only the rudder angle of the OS.

Therefore, this study improves on the authors' previous work (Chun et al., 2021) by proposing a collision avoidance method that considers multiple TSs simultaneously and controls both rudder angle and propeller RPM of the OS.

1.2. Related works

The most recently studied collision avoidance methods for ships use the DRL to generate optimal collision avoidance paths. The DRL performs effectively in collision avoidance problems that require the control of complex dynamic systems in various situations. These can respond rapidly to uncertainty regarding the collision situation. Cheng and Zhang (2018) and Bhopale et al. (2019) proposed a collision

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avoidance method using a Deep Q-Network (DQN; a simple form of the DRL). However, they only considered the static obstacles for collision avoidance. Everett et al. (2018) proposed a collision avoidance method with dynamic obstacles such as a TS that uses long short term memory (LSTM) to generate appropriate avoidance paths even when the number of TSs varies constantly. Cui et al. (2019) combined the DRL with a model-predictive control (MPC) algorithm to generate collision avoidance paths for ships. These studies just considered the distance between the OS and TS, and hard to assess the potential risk of collision. Shen et al. (2019) and Xie et al. (2019) proposed collision avoidance methods for TSs using DQNs. However, owing to the low learning convergence, their application was limited to situations with multiple TSs. Zhao et al. (2019) and Chun et al. (2021) set a TS that was considered the most hazardous even in the presence of multiple TSs, as the input information of the DRL and determined the avoidance path of the OS based on the COLREGs. Sawada et al. (2020) used an LSTM to consider multiple TSs simultaneously and determine an avoidance path based on the COLREGs. Li et al. (2021) proposed a collision avoidance method that combines a DQN and an artificial potential field (APF). Song et al. (2022) proposed a collision avoidance method by combining the visual classification of the CNN with a support vector machine (SVM). Xia et al. (2022) proposed a collision avoidance method using proximal policy optimization (PPO) and a CNN. These are the most commonly used DRLs. Zhai et al. (2022) proposed a collision avoidance method using Double DQN (DDQN), a modified form of DQN. Guan et al. (2023a,b) conducted a collision avoidance study for multiple TSs based on convolutional LSTM which combines the features of the CNN and LSTM. Wang et al. (2023) used approximate representation reinforcement learning (AR-RL) which is based on an approximate representation algorithm to ship collision avoidance. Wu et al. (2023) combined the DRL with the dynamic window approach (DWA) which samples the multiple velocities and calculates the trajectory of the ship. Zheng et al. (2023) applied LSTM to efficient and safe operation for collision avoidance in an environment with multiple TSs. However, these studies controlled only the rudder angle of the OS and not the speed. Therefore, in this study, we propose a collision avoidance method that combines the PPO

and CNN and simultaneously controls the path and speed of the OS by considering the rudder angle and propeller RPM. Also, we apply the proposed method to the Imazu problem (Imazu, 1984), which includes benchmark scenarios of various collision avoidance situations for multiple TSs around the OS, in order to verify its effectiveness. The Imazu problem has been utilized in various studies on collision avoidance for multiple ship encounters, including Sawada et al. (2021), Guan et al. (2023a,b), and Zheng et al. (2023). Table 1 summarizes a comparison between this study and previous studies.

In Table 1, CPA refers to the closest point of approach, which means the closest point when two ships maintain each other's speed. OZT refers to the obstacle zone by target which is a concept that corresponds to a zone that is invaded by a target ship and has a potential risk of collision.

The remainder of this paper is organized as follows. In Section 2, we describe the maneuvering equations for ship motion, collision risk, and various rules for collision avoidance. In Section 3, we propose a DRL-based collision avoidance method. In Section 4, we verify the proposed method and apply it to a complex scenario. Finally, Section 5 concludes the paper and describes future research plans.

2. Mathematical model for collision avoidance

For robust and accurate collision avoidance, it is necessary to accurately simulate the motion of the OS with variations in the rudder angle and propeller RPM. In addition, all ships operating at sea should comply with the COLREGs to prevent collisions between ships. This also needs to be considered while generating collision avoidance paths. Therefore, to make accurate decisions on collision avoidance, this study used a collision risk assessment method that considers the COLREGs and maneuvering performance of the OS (Chun et al., 2021; Ha et al., 2021).

2.1. Maneuvering equations of motion

To accurately simulate the motion of the OS, we used the maneuvering equations of motion of a ship. These are expressed with three degrees of freedom, as shown in Eq. (1): surge, sway, and yaw (Gertler

Table 1
Related works regarding collision avoidance of ship.

Related works	Assessment of the collision risk	Algorithm for the collision avoidance path	Consideration of the COLREGs	Types of obstacles	Consideration of multiple target ships	Control targets
Cheng and Zhang (2018)	No	DQN	No	Static	No	Rudder angle
Everett et al. (2018)	No	LSTM	Yes	Dynamic	Yes	Rudder angle
Bhopale et al. (2019)	No	DQN	No	Static	No	Rudder angle
Cui et al. (2019)	No	MPC, DQN	Yes	Dynamic	No	Rudder angle
Shen et al. (2019)	Yes (Ship domain)	DQN	Yes	Dynamic	No	Rudder angle
Xie et al. (2019)	Yes (CPA)	DQN	Yes	Dynamic	No	Rudder angle
Zhao et al. (2019)	No	PPO	Yes	Dynamic	No	Rudder angle
Zhao and Roh (2019)	No	PPO	Yes	Dynamic	No	Rudder angle
Sawada et al. (2021)	Yes (OZT)	LSTM	Yes	Dynamic	Yes	Rudder angle
Chun et al. (2021)	Yes (Ship domain and CPA)	PPO	Yes	Dynamic	No	Rudder angle
Li et al. (2021)	No	APF, DQN	Yes	Dynamic	Yes	Rudder angle
Song et al. (2022)	No	SVM, CNN	Yes	Dynamic	Yes	Rudder angle
Xia et al. (2022)	Yes (CPA)	PPO	Yes	Dynamic	Yes	Rudder angle
Zhai et al. (2022)	Yes (CPA)	DDQN	Yes	Dynamic	No	Rudder angle
Guan et al. (2023)	Yes (OZT)	LSTM	Yes	Dynamic	Yes	Rudder angle
Wang et al. (2023)	Yes (Ship domain and CPA)	AR-RL	Yes	Dynamic	Yes	Rudder angle
Wu et al. (2023)	Yes (OZT)	DWA, PPO	Yes	Dynamic	Yes	Rudder angle
Zheng et al. (2023)	No	LSTM	Yes	Dynamic	Yes	Rudder angle and propeller RPM
This study	Yes (Ship domain and CPA)	PPO	Yes	Dynamic	Yes	Rudder angle and propeller RPM

and Hagen, 1967).

$$\begin{aligned} m(\dot{u} - vr - x_G r^2) &= X_H + X_P + X_R + X_E \\ m(\dot{v} + ur + x_G r^2) &= Y_H + Y_P + Y_R + Y_E \\ I_{zz}\dot{r} + mx_G(\dot{v} + ur) &= N_H + N_P + N_R + N_E \end{aligned} \quad (1)$$

where m is the mass of the OS, x_G is the center of gravity of the OS in the x-direction, and I_{zz} is the moment of inertia of the OS in the z-direction. u , v , and r are the speed and angular velocity of the OS in the x-, y-, and z-directions, respectively. $\dot{u}$, $\dot{v}$, and $\dot{r}$ are the accelerations with respect to these speeds and angular velocities. X , Y , and N are the fluid forces in the x-, y-, and z-directions and the rotational fluid forces in the z-direction. Each element is divided into the hydrodynamic force (H), propeller force (P), rudder force (R), and environmental disturbance force (E).

Here, the hydrodynamic force is the fluid force acting on the hull of the OS. It was developed by a Mathematical Modeling Group (Kijima and Nakiri, 1990) as Eq. (2):

$$\begin{aligned} X_H &= X_{ii}\dot{u} + X_{vr}vr + X_{rr}r^2 + X_{vv}v^2 + X(u) \\ Y_H &= Y_{iv}\dot{v} + Y_{ir}\dot{r} + Y_{vv}v + Y_{v|v|}|v| + Y_{vrr}vr^2 + Y_{rr}r + Y_{r|r|}|r| + Y_{rvv}rv^2 \\ N_H &= N_{iv}\dot{v} + N_{ir}\dot{r} + N_{vv}v + N_{v|v|}|v| + N_{vrr}vr^2 + N_{rr}r + N_{r|r|}|r| + N_{rvv}rv^2 \end{aligned} \quad (2)$$

Here, the coefficients of velocity (u , v , and r) and acceleration ($\dot{u}$, $\dot{v}$, and $\dot{r}$) X_{ii} , X_{vr} , X_{rr} , X_{vv} , Y_{iv} , Y_{ir} , Y_{vv} , $Y_{v|v|}$, Y_{rr} , $Y_{r|r|}$, Y_{rvv} , Y_{vrr} , N_{iv} , N_{ir} , N_{vv} , $N_{v|v|}$, N_{rr} , $N_{r|r|}$, and N_{rvv} , N_{vrr} are the fluid force coefficients in each direction of the OS. $X(u)$ represents the resistance in a steady sea (Kim and Kim, 2008). In this case, the hydrodynamic force is affected by the current velocity and acceleration of the OS. Therefore, if there is a change in the motion of the OS, the hydrodynamic force will also change continuously. As a result, the speed of the OS can change by the hydrodynamic force without change of the other forces.

The propeller force is the thrust exerted on the OS by the propeller. It mainly acts in the x-direction (surge) of the OS. Therefore, Y_P and N_P can be omitted, and X_P is calculated using Eq. (3):

$$X_P = \rho n^2 D_p^4 K_T(J) \quad (3)$$

where ρ is the seawater density and n is the propeller RPM. D_p is the diameter of the propeller, and $K_T(J)$ is the propeller coefficient according to J . Here, J is defined as in Eq. (4):

$$J = \frac{u}{nD_p} \quad (4)$$

The rudder force refers to the fluid force exerted on the OS by the rudder and can be expressed as Eq. (5):

$$\begin{aligned} X_R &= X_\delta \delta \\ Y_R &= Y_\delta \delta \\ N_R &= N_\delta \delta \end{aligned} \quad (5)$$

where X_δ , Y_δ , and N_δ are the rudder coefficients calculated considering the rudder area and the coefficients for the additional drag.

Finally, the OS is affected by various external forces such as wind, waves, and currents during operation. The force acting on the OS is expressed as an environmental disturbance force.

2.2. COLREGs regulations

Rules 12–17 of the COLREGs categorize ship-to-ship encounter situations according to the relative position and direction of each ship. These also describe the rules for the direction of travel of the ships in each situation. At this time, the ship that is obligated to avoid the other ship and perform collision avoidance is called the “give-way ship”, and the ship that is obligated to maintain the current path is called the “stand-on ship”. In each encounter, the OS can be a give-way ship or stand-on ship to the TS.

The COLREGs specify four types of encounter situations between ships (Chun et al., 2021; Zhao, 2019; Zhao and Roh, 2019). The head-on situation refers to a situation in which the OS and TS face and approach

each other. The crossing give way situation occurs when the TS approaches the right side of the OS. An overtaking situation occurs when the OS overtakes the TS from its rear. Finally, a crossing stand on situation occurs when the TS approaches the left side of the OS. The give-way ship, stand-on ship, and direction of travel of the give-way ship for each encounter according to the COLREGs are listed in Table 2.

In addition, the COLREGs Rule 8(b) states that the give-way ship should clarify to the stand-on ship that it is performing collision avoidance and should avoid successive alterations.

2.3. Collision risk assessment

For an autonomous navigation system to accurately determine the potential collision risk of TSs around an OS and perform collision avoidance accordingly, it is necessary to quantitatively assess the collision risk for each TS. In this study, the collision risk assessment method proposed in our previous study (Chun et al., 2021; Ha et al., 2021) was used. In that study, the collision risk was calculated based on the ship domain method to ensure reliable collision avoidance. Then, the quantitative collision risk was calculated based on the CPA method. The CPA method is generally used for collision risk assessment and collision avoidance with the CPA-based indices (Zhou et al., 2021; Kim et al., 2022; Seo et al., 2023).

First, the ship domain area is defined as an asymmetric ellipse shape with the front and right lengths determined as L_L (long length of the ship domain) and the rear and left lengths determined as L_S (short length of the ship domain). LL and LS are set as the minimum distances wherein the OS can avoid the TS according to the athletic performance of the OS.

In this study, Eq. (6) was used to calculate the collision risk. The collision risk is calculated using two coefficients: the time to CPA (TCPA) and distance to CPA (DCPA). Then, the coefficients are adjusted through an exponential function so that the collision risk becomes one at the boundary line of the OS's ship domain area. The value reduces as the TS moves away from the OS and becomes zero when the risk disappears completely. That is, the collision risk is hazardous as it increases from zero to one. A value above one indicates that the OS and TS collide.

$$CR = f_{angle} \exp(-DCPA/a) \cdot \exp(-TCPA/b) \quad (6)$$

In this case, the collision risk factor f_{angle} normally has a value of one. However, it has a value of 0.25 to prevent the collision risk from being assessed as more hazardous than that when the TCPA < 0, i.e., when the OS and TS are moving away from each other. Without f_{angle} , the collision risk is excessively high compared to the actual threat even though the TS is not a significant threat to the OS when the OS and TS are actually moving away.

3. Collision avoidance based on the DRL with path-speed control

The DRL is a branch of machine learning inspired by behavioral psychology based on the Markov decision process (MDP). It mathematically models the decision-making process and uses an artificial neural network of deep learning as the decision-maker (Sutton and

Table 2

Give-way ship, stand-on ship, and direction of give-way ship.

	Own ship	Direction of own ship	Target ship	Direction of the target ship
Head-on	Give-way	Right	Give-way	Right
Crossing give way	Give-way	Right	Stand-on	Straight
Overtaking	Give-way	Right or Left	Stand-on	Straight
Crossing stand on	Stand-on	Straight	Give-way	Right

Barto, 2018). Reinforcement learning is a method for identifying an optimal “policy” that selects a sequence of actions that maximizes a “reward” among the available “actions” by identifying the current “state” of an entity “agent” that corresponds to a decision-maker defined in a certain “environment” (Chun et al., 2021, 2022; Zhao, 2019; Zhao and Roh, 2019; Bae et al., 2022; Oh et al., 2022; Kim et al., 2023; Kong et al., 2023). Fig. 1 shows the overview of the process of the DRL.

In Fig. 1, the environment calculates and delivers the next state and reward by the current state and the action derived from the agent. Here, the state is the information that serves as a basis for the agent to select the next action, and the reward is the score that evaluates the result of the action derived from the agent. The agent receives state and reward from the environment and derives the next action by policy. Policy is a probability distribution to select action, from the given state. Action is derived from the policy and delivered to the environment to calculate the next state and reward. At last, the return is a discounted sum of rewards in a scenario. The DRL is the learning process of maximizing expected return.

By this method, we can identify potential hazards to the TS in advance, prevent hazardous situations, and respond flexibly to unforeseen situations. Moreover, it can achieve more realistic collision avoidance by simultaneously controlling the path and speed of the OS to generate a collision avoidance path. In this paper, we propose a collision avoidance method that learns a collision avoidance model based on the PPO algorithm (Long et al., 2018; Schulman et al., 2017) during the DRL.

3.1. Formulation of the state

The agent selects the next action based on the state. The environment calculates the reward based on the state modified by the action to determine whether the selected action is appropriate. At this time, the agent selects the next action through the policy. The DRL policy is composed of an artificial neural network that adopts the state as the input and calculates the action as the output. The DRL is based on MDP, which states that all the elements of a state should be observable and consist of values calculated from the previous state. In most studies, the state is represented by a 1D multi-layer perceptron (MLP). Therefore, it cannot be represented by the state when the number of TSs varies dynamically. That is, multiple TSs cannot be considered simultaneously, and only the TS with the highest collision risk is considered to be the

state. Therefore, in this paper, we propose a collision avoidance method using the MLP-CNN state. It is a combination of an MLP state consisting of elements for the OS, destination, route, etc., and a CNN state representing the information of the TSs within a certain distance around the OS in a two-dimensional grid. First, the MLP state s_{mlp}^t is constructed as illustrated in Eq. (7):

$$s_{mlp}^t = [v_{Ox} \ v_{Oy} \ ||\mathbf{P}_{goal} - \mathbf{P}_O|| \ \delta \ y_e \ \psi_e \ \chi_e \ \dot{\psi}_e \ \dot{\chi}_e \ \tilde{\varphi} \ n \ CR] \quad (7)$$

In Eq. (7), the state s^{tO} associated with the OS consists of 11 elements: v_{Ox} and v_{Oy} are the speeds (x- and y-directions) of the OS in a two-dimensional plane, and $||\mathbf{P}_{goal} - \mathbf{P}_O||$ represents the distance between the current position of the OS and the destination. δ is the rudder angle of the OS. y_e is the difference between the current position of the OS and the reference path. ψ_e and χ_e represent the heading angle error and course angle error, and $\dot{\psi}_e$ and $\dot{\chi}_e$ represent their derivatives with respect to time. $\tilde{\varphi}$ is the relative angle between the course angle of the OS and the angle directed toward the destination from this ship. n is the propeller RPM of the current OS, and CR is the collision risk of the TS with the highest collision risk among the TSs around the OS. Each value of the MLP state is scaled to a minimum value between zero and one.

In addition, this study converted the information of the TSs within a certain distance of the OS into a two-dimensional grid and defined the CNN state based on the grid. The configuration of the CNN is illustrated in Fig. 2.

In Fig. 2, a square centered on the OS and within a certain distance D along the x- and y-axes is input as a state. Here, D is the observation distance. Furthermore, the square is divided into N equal parts in each axis direction. A 2D vector with a total size of $N \times N$ is defined as the CNN state. Here, N denotes the number of cells. Each cell in the CNN state is initialized with a value of zero. When each TS enters the corresponding CNN state, the collision risk of the TS is set to the value of the cells in the CNN state occupied by the TS according to the position of the TS. Thus, the CNN state can have a value of at least zero and at most one. The CNN performs a convolution process that goes through a convolutional layer and is commonly used in image processing. It is utilized to reduce the computation of the MLP based on the fully-connected layer. If the $N \times N$ 2D vector for the TS is learned based on the general MLP, the required computation is excessively increased and it is difficult to converge normally. Therefore, in this study, MLP-CNN state is proposed

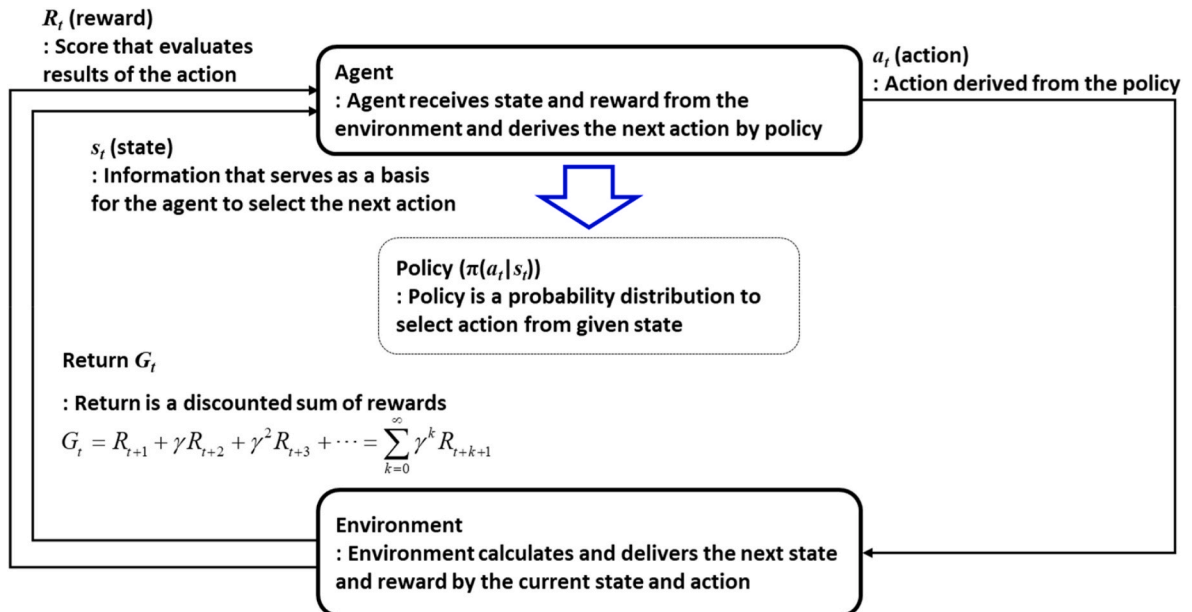


Fig. 1. Overview of the process of the DRL.

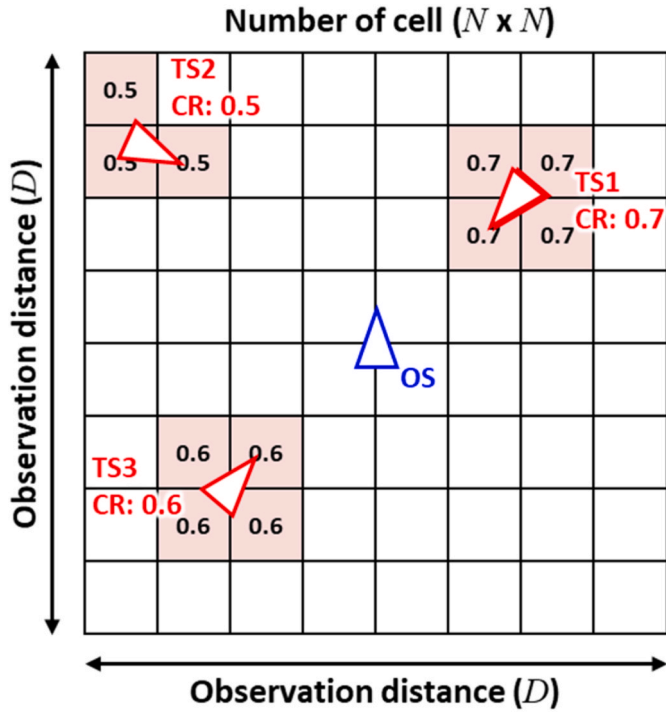


Fig. 2. Definition of the CNN state.

to reduce the amount of computation based on the convolutional layer. The proposed collision avoidance method adopts as input an MLP-CNN state that simultaneously uses a 1D input of the MLP state with 12 and a 2D input of a CNN state with $N \times N$.

3.2. Formulation of the action

The agent selects an action from the current state based on the policy. The environment calculates the next state and reward based on the selected action. In a complex route, the navigator should adjust the path and speed of the OS to ensure safety. Moreover, it is necessary to make appropriate decisions. In this study, the rudder angle and propeller RPM of the OS were set to control its path and speed. The rudder angle of the OS is determined as a value between $-\delta_{\max}$ and $\delta_{\max}$ depending on the ship type and maximum angle by which the rudder can be turned ($\delta_{\max}$). The propeller RPM of the OS is similarly determined as a value between zero and $n_{\max}$ depending on the ship type and propeller characteristics. In this case, $n_{\max}$ is the maximum value of the propeller RPM.

The collision avoidance method proposed in this study consists of two values, as shown in Eq. (8):

$$a' = [\delta \quad n] \quad (8)$$

where δ is the rudder angle of the OS and has a value between $-\delta_{\max}$ and $\delta_{\max}$. n is the propeller speed and has a value between zero and $n_{\max}$.

At this time, the hyperbolic tangent function is used as the activation function of the output of the neural network applied in this study, and the function has a value between -1 and 1 . Therefore, in this study, the rudder angle is applied as a value between $-\delta_{\max}$ and $\delta_{\max}$, and the propeller RPM is applied as a value between zero and $n_{\max}$ through scale change for the value between -1 and 1 . Also, if the rudder angle and propeller RPM suddenly change significantly, it is possible to perform unrealistic control that ignores the actual control response of the rudder and propeller. Therefore, in this study, the rudder angle and propeller RPM are limited by δ_{step} and n_{step} to the values that can be changed in a time step to prevent an unrealistic control.

The rudder angle and propeller RPM are used to calculate the rudder force and propeller force (defined in Section 2.1), respectively. These

control the path and speed of the OS.

3.3. Formulation of the rewards

In this study, we proposed a collision avoidance method using the DRL based on the actor-critic method (Chun et al., 2021; Zhao, 2019; Zhao and Roh, 2019). The actor network derives the next action using the state received from the environment, and the critic network evaluates the suitability of the action using the value function calculated from the reward. Therefore, the reward is defined based on an evaluation metric for the agent to learn to select the correct action in each situation. In this study, it was defined as a function for the OS to efficiently arrive at the final destination by avoiding hazardous situations. We divided the rewards into the efficiency reward (R_e ; it evaluates whether the OS reaches the destination at the desired target speed along the reference path (a straight line connecting the starting point and destination)) and safety reward (R_s ; it indicates that the OS avoids hazardous situations). The total reward for each time step is calculated as the sum of the efficiency and safety rewards, as shown in Fig. 3.

In Fig. 3, the efficiency reward consists of the goal, cross, and speed rewards, and the safety reward consists of the risk, COLREGs, and heading rewards. In this study, all the rewards had a value between -1 and 1 to equalize each of their influence. Each reward is described below.

3.3.1. Efficiency reward

As mentioned previously, the efficiency reward indicates that the OS reaches its destination along the desired path at the target speed. The efficiency reward is divided into the goal reward based on the distance from the final destination, the cross reward based on the distance from the reference path, and the speed reward based on the difference between the target speed and current speed of the OS.

3.3.1.1. Goal reward. Goal reward (R_{goal}) refers to the reward given based on the distance between the position of the OS and the destination ("goal"). It is calculated based on the proximity of the current position to the destination based on the previous position of the OS.

As shown in Fig. 4, the OS travels to the next position based on the variation in speed and path from the previous position ($\mathbf{P}_O^{t-1}$). At this time, based on the speed (v_O) of the OS, the current position $\mathbf{P}_O^t$ would be within a circle with a radius of $v_O \cdot dt$ and centered on $\mathbf{P}_O^{t-1}$ (see Fig. 4). Here, dt denotes the time step. When the OS is located at Point A in Fig. 4, it is closest to the destination. When it is located at Point B in Fig. 4, it is farthest from the destination. In this study, to ensure that the goal reward has a value of one when the OS is closest to the destination (Point A in Fig. 4) and -1 when it is farthest away (Point B in Fig. 4), we defined the goal reward R_{goal} as in Eq. (9):

$$R_{\text{goal}} = \frac{D_{\text{goal}}^{t-1} - D_{\text{goal}}^t}{v_O \cdot dt}, D_{\text{goal}}^t = \|\mathbf{P}_{\text{goal}} - \mathbf{P}_O^t\| \quad (9)$$

where D_{goal}^{t-1} and D_{goal}^t the distances between the destination and the previous and current positions of the OS, respectively. $\mathbf{P}_{\text{goal}}$ is the location of the destination. $\mathbf{P}_O^t$ is the current position of the OS. v_O is the speed of the OS, and dt is the time step.

3.3.1.2. Cross reward. The cross-reward (R_{cross}) is determined by the distance between the position of the OS and reference path. Here, the reference path represents the optimal path from the origin to the destination. If a reference path is not provided, it is defined as the straight-line from the origin to the destination.

In Fig. 5, d_{cross}^t is the distance between the position of the OS and reference path. To avoid a collision between the OS and TS, it is necessary to reduce the collision risk of the TS even if it deviates from the reference path (which may increase d_{cross}^t). However, the smaller d_{cross}^t is, the closer is the OS to the reference path, and the shorter is the

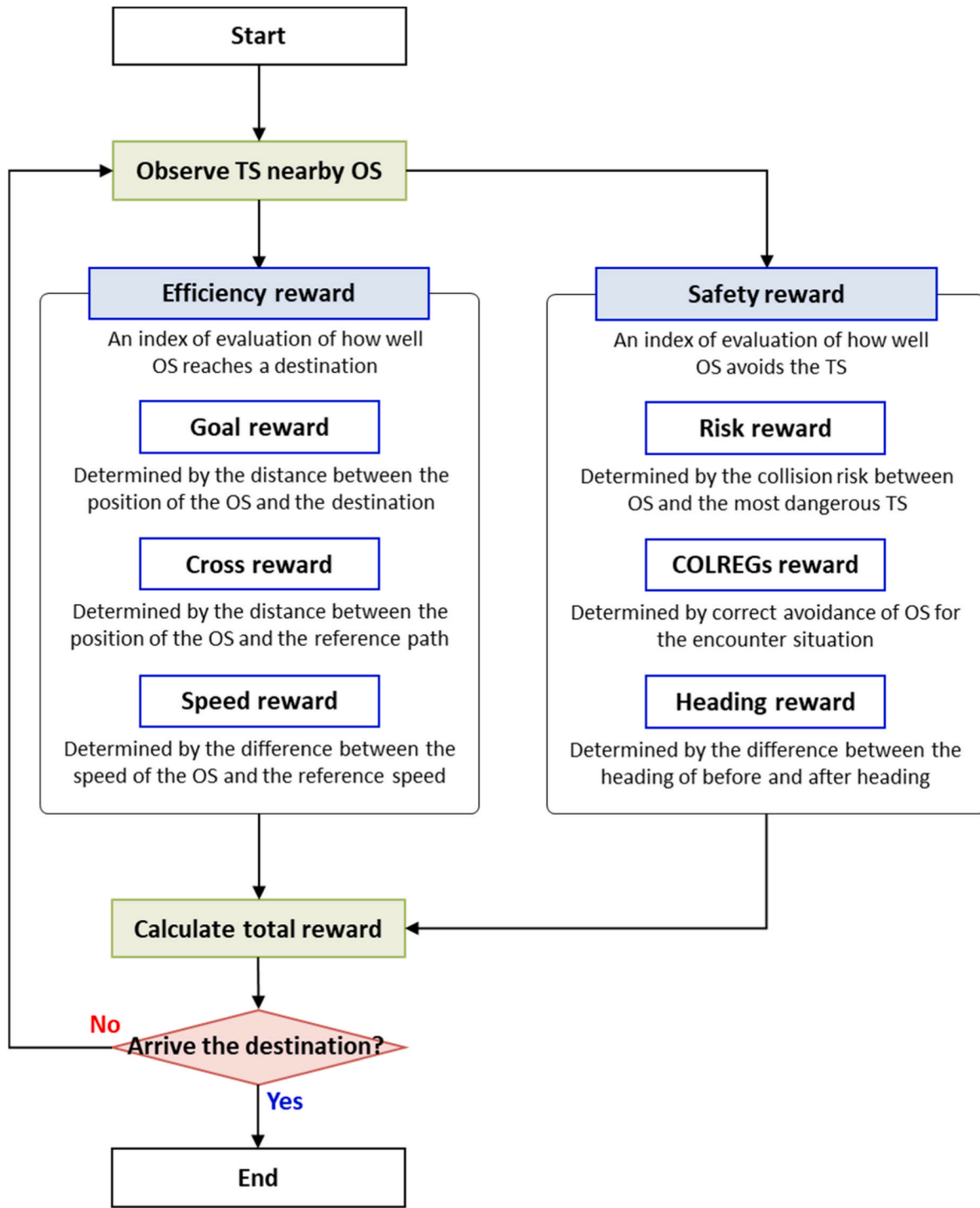


Fig. 3. Procedure for calculating the total reward.

route to the destination. In addition, $d_{\max}$ is the maximum distance, i.e., the maximum value of d_{cross}^t . If the OS exceeds this distance, the scenario is assumed to have failed. Therefore, from the perspective of the efficiency reward, the smaller d_{cross}^t is, the more efficient is the operation. Accordingly, the cross-reward R_{cross} is defined by Eq. (10):

$$R_{\text{cross}} = \frac{(d_{\max} - 2 \cdot d_{\text{cross}}^t)}{d_{\max}} \quad (10)$$

If the OS is on the reference path, d_{cross}^t has a value of zero. In this case, the cross-reward is one. Conversely, if the OS is farther away from the reference path by $d_{\max}$ (i.e., the scenario fails), the cross reward is

−1.

3.3.1.3. Speed reward. The speed reward (R_{speed}) is determined by the target and current speeds of the OS. Here, the target speed is the speed at which the most efficient operation can be achieved according to the fuel efficiency of the main engine of the OS and the complexity of the route. If the target speed is not provided, the service speed of the OS is defined as the target speed. In this study, when the current speed of the OS was within the target speed, the speed reward had a maximum value of one. Meanwhile, if the current speed of the OS was zero or more than two times the target speed, the speed reward had a minimum value of −1. In

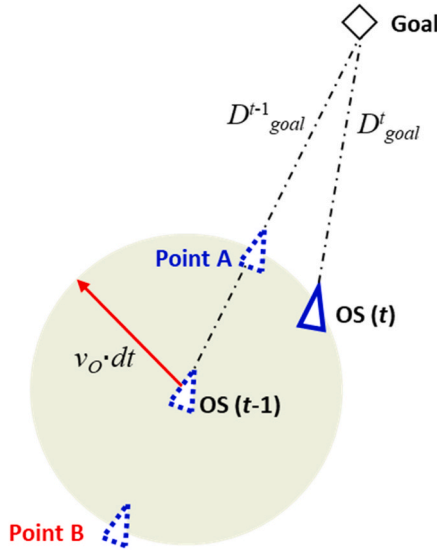


Fig. 4. Definition of the variables for the goal reward.

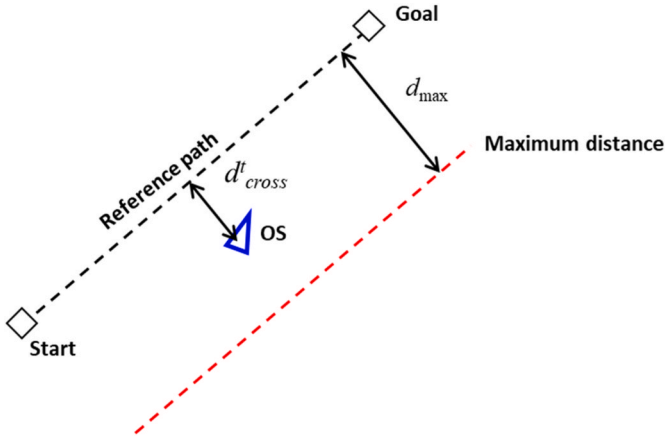


Fig. 5. Definition of the variables for the cross reward.

addition, the speed reward was set linearly between these values. Accordingly, the speed reward was defined by Eq. (11):

$$R_{\text{speed}} = \begin{cases} \frac{2v_O - v_{\text{ref}}}{v_{\text{ref}}} & \text{if } v_O \leq v_{\text{ref}} \\ \frac{3v_{\text{ref}} - 2v_O}{v_{\text{ref}}} & \text{if } v_O > v_{\text{ref}} \end{cases} \quad (11)$$

where v_O is the current speed of the OS and v_{ref} is the target speed.

3.3.2. Safety reward

The safety reward evaluates whether the OS can avoid the TSs and perform safe operations in hazardous situations. The safety reward is divided into the risk reward based on the collision risk of the most hazardous TS, the COLREGs reward based on compliance with rules 12–17 for avoidance situations, and the heading reward based on the variation in the heading angle of the OS.

3.3.2.1. Risk reward. The risk reward (R_{risk}) is determined by the value of the collision risk between the OS and the TS with the highest collision risk (the most hazardous TS). The collision risk was calculated according to the method described in Section 2.3. As mentioned earlier, a collision risk of zero implies that the OS is in a safe situation with no hazards. Meanwhile, a collision risk of one implies that the OS and TS have

collided. Therefore, when the collision risk is zero, the risk reward is calculated as one. In addition, in a situation where the OS and TS collide, the risk reward is calculated as -1 . In this case, the collision risk is given a margin of $CR_{\text{allowable}}$. Therefore, if the collision risk is less than $CR_{\text{allowable}}$, the risk reward is fixed at one. Meanwhile, if the collision risk is higher than $CR_{\text{allowable}}$, the risk reward decreases linearly until the collision risk is equal to one. This results in a risk reward of -1 for a collision risk of 1. That is, $CR_{\text{allowable}}$ can be a criterion for the OS to initiate collision avoidance with the TS. The risk reward R_{risk} is defined by Eq. (12):

$$R_{\text{risk}} = \begin{cases} 1 & \text{if } CR \leq CR_{\text{allowable}} \\ -\frac{2}{1 - CR_{\text{allowable}}} CR + \frac{1 + CR_{\text{allowable}}}{1 - CR_{\text{allowable}}} & \text{if } CR > CR_{\text{allowable}} \end{cases} \quad (12)$$

$CR_{\text{allowable}}$ can be a criterion for the OS to initiate collision avoidance with the TS. CR is the collision risk with the most hazardous TS.

3.3.2.2. COLREGs reward. The COLREGs reward (R_{colregs}) is determined by the correct collision avoidance performed in the four encounter situations, according to the COLREGs rules 12–17 described in Section 2.2. In this study, we calculated the COLREGs reward as one if the TS's collision risk was lower than the $CR_{\text{allowable}}$ defined earlier for the risk reward (safe situation) or if the OS undertook a rule-compliant move in a collision avoidance situation where the collision risk was higher than $CR_{\text{allowable}}$. Conversely, in a collision avoidance situation, if the OS did not travel according to the rules for the TS, we calculated the COLREGs reward as -1 . The formula for the COLREGs reward is given by Eq. (13):

$$R_{\text{colregs}} = \begin{cases} 1 & \text{if } CR \leq CR_{\text{allowable}} \text{ or satisfies rules} \\ -1 & \text{otherwise} \end{cases} \quad (13)$$

3.3.2.3. Heading reward. The heading reward (R_{heading}) is determined according to Rule 8(b) mentioned in Section 2.2. This is to ensure that the OS clearly expresses the collision avoidance path to the TS in a collision avoidance situation and does not cause ambiguity, such as frequent heading alterations. In this study, we calculated the heading reward based on the variation in the OS's heading angle from the previous time. That is, if the heading angle does not vary, the heading reward would be one. Conversely, the heading reward decreases linearly as the variation in the heading angle of the OS increases. If the variation in the OS heading angle is larger than φ_{max} (the maximum shift value), the heading reward becomes -1 . The heading reward can be expressed using Eq. (14):

$$R_{\text{heading}} = \begin{cases} 1 & \text{if } \varphi_{\text{diff}} = 0 \\ 1 - 2 \frac{\varphi_{\text{diff}}}{\varphi_{\text{max}}} & \text{if } 0 < \varphi_{\text{diff}} \leq \varphi_{\text{max}} \\ -1 & \text{if } \varphi_{\text{diff}} > \varphi_{\text{max}} \end{cases} \quad (14)$$

where $\varphi_{\text{diff}} = |\varphi_{\text{before}} - \varphi_{\text{after}}|$. φ_{before} , φ_{after} , and φ_{diff} are the previous heading angle of the OS, its current heading angle, and their difference, respectively. φ_{max} is the maximum shift in the OS's heading angle.

4. Applications

In this study, we proposed a method to comprehend the surrounding situation based on an MLP-CNN, calculate the collision risk for each TS, and perform collision avoidance simultaneously in a situation where an OS and multiple TSs are encountered. To verify the effectiveness of the proposed method, a 58:1 scale model ship of KVLCC2 (Kim and Kim, 2008) was used as an OS to learn the collision avoidance method and was applied to an example. Table 3 summarizes the main specifications and used for the validation.

Also, the maneuvering coefficients of the OS described in Eq. (2) are summarized in Table 4.

Table 3

Principal dimensions and key parameters of the own ship in this example.

Item	Value	Item	Value
Length	5.5172 m	Breadth	1 m
Depth	0.5172 m	Target speed (v_{ref})	1 m/s
Short length of ship domain (L_S)	5.5 m	Long length of ship domain (L_L)	15.5 m
Maximum rudder angle	15°	Maximum propeller RPM	600
δ_{step}	5°	n_{step}	120
$CR_{\text{allowable}}$	0.3		

Table 4

Maneuvering coefficients (Kijima and Nakiri, 1990) of the own ship in this example.

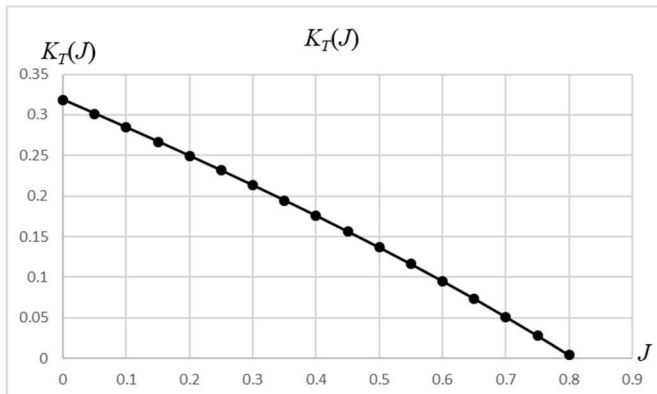
Item	Value	Item	Value
X'_{il}	-0.000954	X'_{vr}	0.001688
X'_{rr}	0.001219	X'_{vv}	0.0013821
$Y'_{\dot{v}}$	-0.015104	Y'_r	-0.001428
Y'_v	-0.016190	$Y'_{v v }$	-0.030441
Y'_r	0.004720	$Y'_{r r }$	0.000205
Y'_{rvv}	0.021775	Y'_{vrr}	-0.022999
$N'_{\dot{v}}$	-0.000785	N'_r	-0.000800
N'_v	-0.008754	$N'_{v v }$	0.001250
N'_r	-0.003115	$N'_{r r }$	0.000038
N'_{rvv}	-0.0017563	N'_{vrr}	0.001063

The OS's propeller is a single propeller with a diameter of 0.17 m and four blades. Furthermore, the relationship between the advance ratio (J) and thrust coefficient $K_T(J)$ for the OS's propeller described in Section 2.1 is shown in Fig. 6.

As shown in Fig. 6, the thrust coefficient K_T has a maximum value of 0.3183 when J is zero. It then decreases gradually to zero when J is 0.8.

In this study, we applied the proposed method to the Imazu problem (1984). Twenty benchmark scenarios were generated by varying the combinations of each ship's encounter situations. It is designed as an example of a combination of four 1:1 ship encounter situations simplified by the COLREGs, with each encounter situation varied by several angles and directions. The Imazu problem has been utilized in various studies on collision avoidance for multiple ship encounters, including Sawada et al. (2021), Guan et al. (2023a,b), and Zheng et al. (2023). Fig. 7 shows each scenario for the Imazu problem.

In this case, the start position of the OS is (0, -100), and the destination position is (0, 100). Here, the x- and y-coordinates of each position are in meters. In the Imazu problem, the TS is categorized as head-on, give way, overtaking, stand on, southeast, southwest, south-south-

**Fig. 6.** Thrust coefficient of the propeller of own ship.

southeast, south-southeast, south-southwest, northeast, etc., according to the starting position and heading angle. In all the examples, the position, heading angle, and speed of the TS according to each classification are listed in Table 5. These were varied randomly within a certain threshold while training the DRL-based collision avoidance model.

4.1. Performance test according to the level of learning

The proposed collision avoidance method receives feedback based on the total reward, which is used to learn how efficiently the OS avoids the TSs and reaches the destination. In this process, the DRL-based collision avoidance model selects actions with an increasingly high total reward as it learns. Thereby, it avoids hazardous situations and reaches the destination via an efficient route. In this section, we compare and analyze the performance of the DRL-based collision avoidance models according to their level of learning.

As shown in Table 6, when learning progressed negligibly (10 iterations), the OS did not attempt to avoid collisions with the TS. This resulted in collisions in most scenarios. Therefore, the average value of the maximum collision risk of the TS for all the scenarios exceeded one and that for the minimum distance between the OS and TS was measured to be approximately 12.8 m. In addition, the average speed of the OS was 0.63 m/s. However, as the learning of the DRL-based collision avoidance model progressed gradually, the number of successful scenarios increased gradually. When the learning was completed, collision avoidance was successful for all the scenarios. In this case, the average maximum collision risk of the TS was 0.94 (which was the lowest), and the average speed of the OS was 0.79 m/s (which was the closest to the target speed). The average speed of the OS was marginally lower than the target speed because the OS temporarily decreased its speed while avoiding the TS and tended to increase its speed again after the TS had passed completely. Fig. 8 shows the collision avoidance path of the OS as a function of the learning level in the four scenarios of the Imazu problem.

Fig. 8 shows scenarios 7, 14, 17, and 20 of the Imazu problem. Here, the collision avoidance paths differ significantly depending on the level of learning and are suitable for a comparative analysis. First, in scenarios 7, 14, and 20, before learning is complete, the OS collides because it fails to avoid a TS while avoiding another. In addition, in scenario 17, although the OS avoids collisions and reaches the destination, it travels on a relatively risky route. This results in a high collision risk (0.99). In contrast, the OS with the trained model avoids collisions with TSs in all the scenarios and reaches the destination (particularly in scenarios with dense TSs, such as scenario 17) by selecting a safer route with a lower density of TSs (which reduces the collision risk (0.93)). We also observed that in most scenarios, the OS travels to the starboard to comply with the COLREGs rules. Moreover, certain OSs travel temporarily to the port to avoid hazardous situations when the TSs are clustered on the starboard side. Fig. 9 shows the speed of the OS for each scenario in Fig. 8.

In each scenario, the OS begins at a speed of 0 m/s and has a target speed of 1 m/s. Moreover, the speed of the OS initially increases gradually and reduces temporarily when a collision avoidance maneuver requires a turn or when a nearby TS endangers the OS. After the OS is out of a hazardous situation and its path is stabilized, it regains speed and approaches the target speed.

4.2. Performance test according to parameter tuning

A collision avoidance model based on the MLP-CNN state is proposed in Section 3.1 to identify multiple TSs simultaneously. In the process of defining the CNN state to comprehend the surrounding situation, two parameters are defined: the observation distance (D) and number of cells (N). These also affect the cell size (l_c) of the CNN state, which is calculated using Eq. (15):

$$l_c = D/N \quad (15)$$

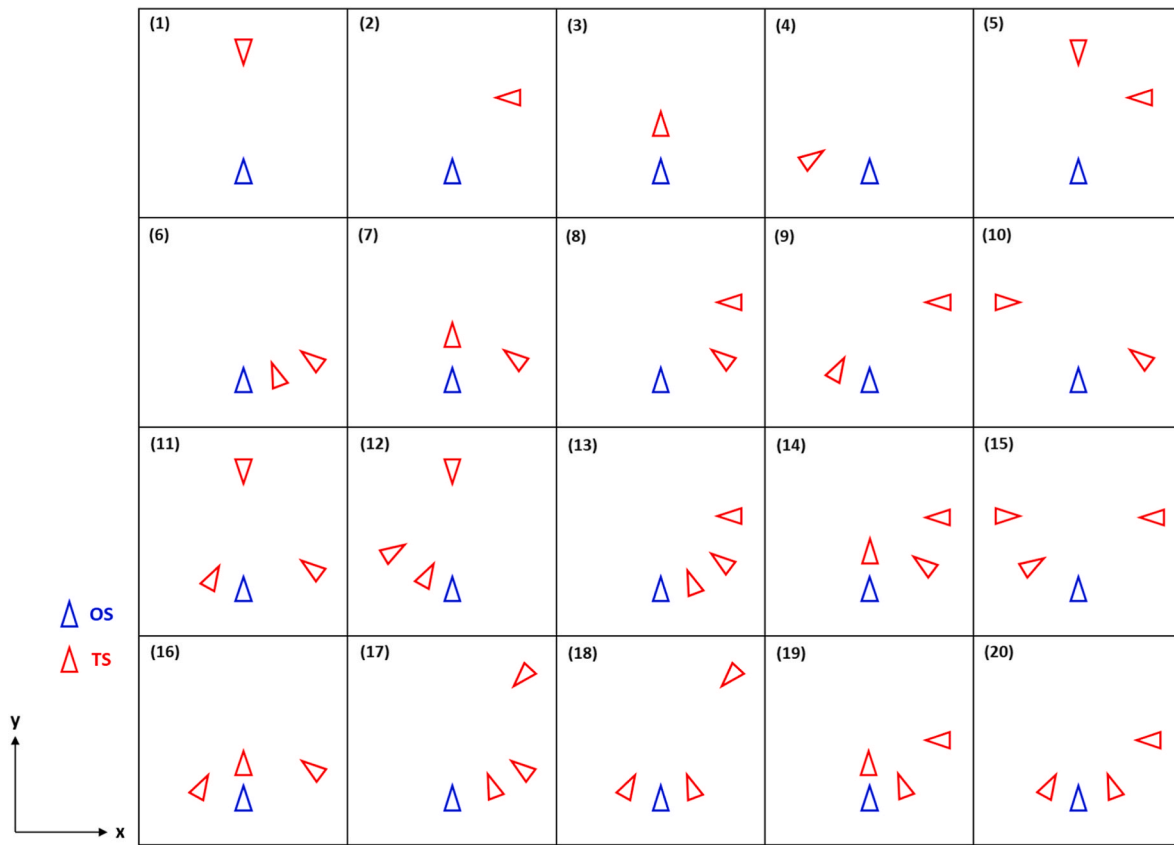


Fig. 7. 20 benchmark scenarios of the Imazu problem.

Table 5

TS information of each situation.

	x [m]	y [m]	Heading angle [°]	Speed [m/s]
Head on	0	75	−90	1
Give way	75	0	180	0.5
Overtaking	0	−50	90	0.1
Stand on	−75	0	0	0.5
Southeast	75	75	135	0.5
Southwest	−75	−75	45	0.5
South-southeast	50	−100	105	0.5
South-southwest	−50	−100	75	0.5
Northeast	75	75	225	0.9

Table 6

Result of collision avoidance according to the level of learning.

	Number of success scenarios	Average maximum collision risk of the TS	Average minimum distance of the TS [m]	Average speed [m/s]
10 iterations	2/20	1.02	19.8	0.63
100 iterations	4/20	1.02	20.4	0.68
1000 iterations	11/20	0.98	23.3	0.74
10,000 iterations (learning completed)	20/20	0.94	25.2	0.79

The cell size calculated in this manner is the basis for identifying the position and shape of the TS. The smaller the cell size, the more accurate is the TS. This implies that the larger the number of small pieces into which the TS of an equal size is divided, the more accurate is the shape comprehension. Moreover, it has an effect similar to that when the

resolution is increased. To reduce the cell size, the observation distance should be decreased or the number of cells should be increased. This is shown in Fig. 10.

In this case, if the observation distance decreases, the cell size decreases. However, the disadvantage is that the OS does not identify nearby TSs because the area it evaluates becomes narrower. Meanwhile, as the number of cells increases, the cell size decreases because the cells are divided more tightly. However, the amount of computation increases exponentially, thereby resulting in a long learning time and the difficulty of applying it in real time. Therefore, in this study, we compared the learning results for each parameter variation and evaluated the avoidance performance, as listed in Table 7.

The collision avoidance path as the observation distance varies is shown in Fig. 11.

In Fig. 11, we can observe that as the observation distance increases, the average collision avoidance path becomes more distant because the potential threat from the TS is identified in advance. Furthermore, there are relatively few abrupt path alterations because the TS is observed from a distance. However, because the resolution of the surroundings decreases as the cell size increases and it may be difficult to identify the surrounding situation accurately, there are scenarios (9, 13 for 500 m and 7, 14 for 1000 m) in which collision avoidance fails in a dense situation with TSs when the observation distance is 500 m and 1000 m, respectively. This is also related to the collision risk calculation method used in this study. It defines $CR_{allowable}$ as 0.3 and fixes the risk reward for a given TS to one if the collision risk is less than $CR_{allowable}$. Therefore, even if a TS is observed from a distance, collision avoidance is not performed if the collision risk is below $CR_{allowable}$. Therefore, it is important to set an appropriate observation distance for the OS, TS, and collision risk. Table 8 presents the collision avoidance result as a function of the observation distance.

In Table 8, when the observation distances are 100 m and 250 m,

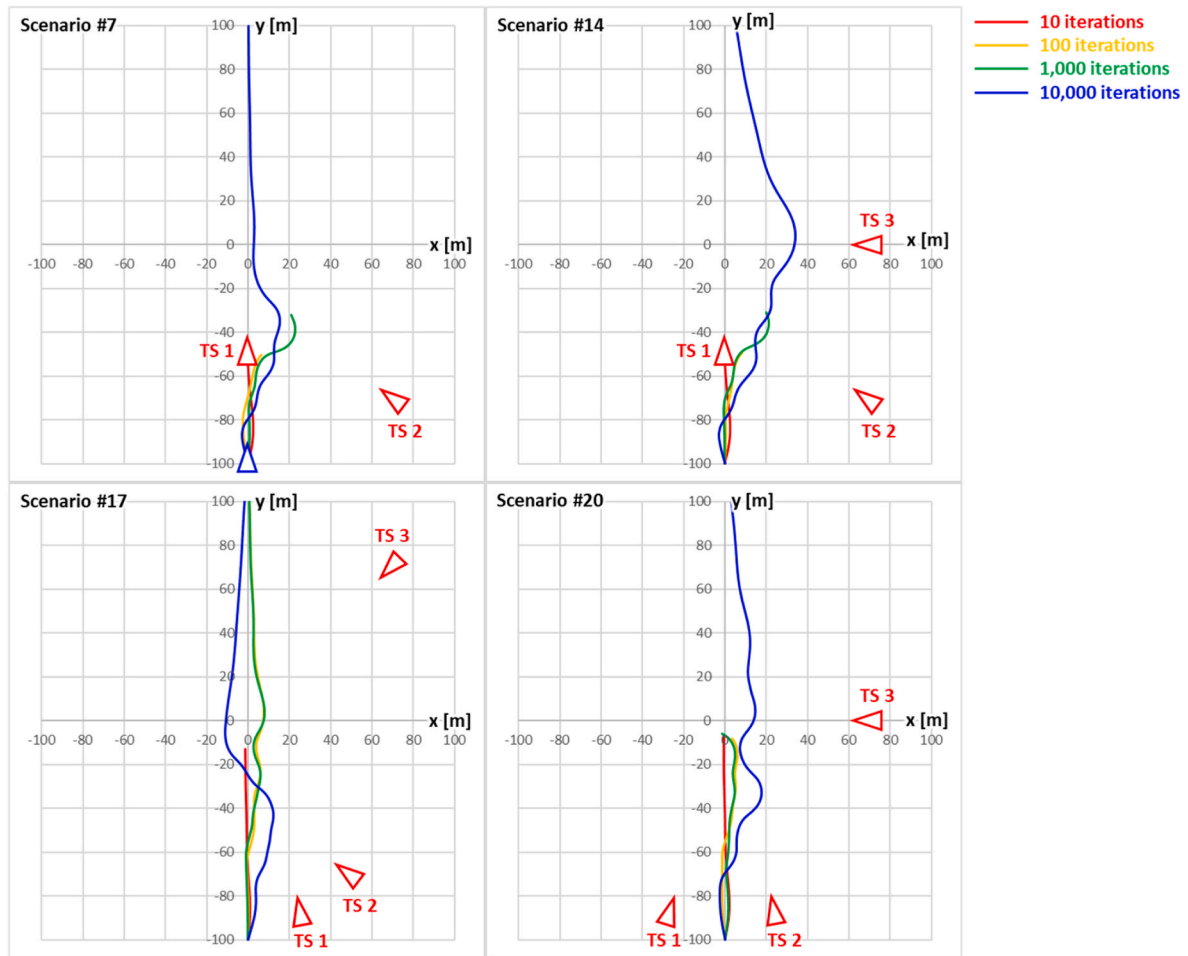


Fig. 8. Collision avoidance path for the representative scenarios according to the level of learning.

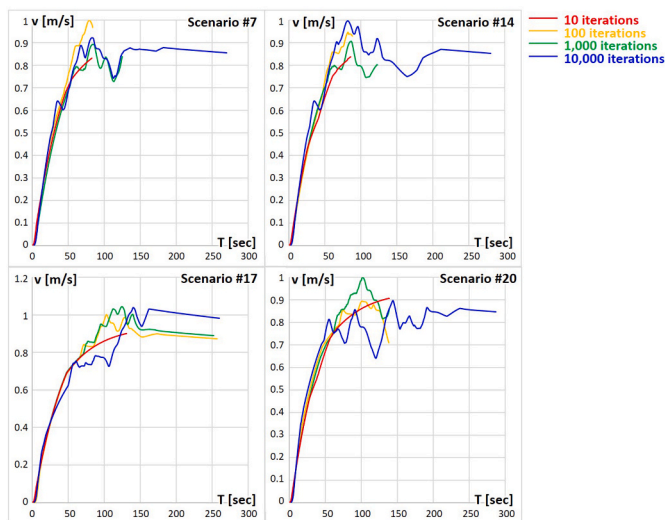


Fig. 9. Speed of the OS according to the level of learning.

collision avoidance is successful for all the scenarios of the Imazu problem. When the observation distance is 100 m, the collision avoidance route is more efficient in terms of the average speed, travel time, and travel distance, although the collision risk is marginally higher at 0.88. In contrast, an observation distance of 250 m generates a safer collision avoidance route with a lower collision risk of 0.87.

Additionally, when the observation distance is 500 m and 1000 m, there are scenarios (9, 13 for 500 m and 7, 14 for 1000 m) in which collision avoidance fails owing to the factors described above.

The collision avoidance paths based on the number of cells are shown in Fig. 12.

As shown in Fig. 12 as the number of cells decreases, the resolution of the surroundings decreases because the cell size increases. This hinders the accurate perception of the surroundings. Therefore, when the number of cells is 50 and 100, collision avoidance fails in several scenarios (8, 9, 15, and 20 for 50, and 7 and 14 for 100). In particular, when the number of cells is 50, collision avoidance fails in several scenarios (8, 9, 15, and 20). In addition, as the number of cells increases, the surrounding situation can be identified accurately, and the amount of computation increases exponentially because of the calculation of the association between the cell and surrounding TSs and the training of the collision avoidance model. Thus, the learning does not converge normally, and collision avoidance fails in most scenarios (5, 7, 8, 9, 12, 14, 15, 19, and 20) when the number of cells is 500. Table 9 presents the results of collision avoidance based on the number of cells.

In Table 9, the only case in which collision avoidance is successful for all the scenarios of the Imazu problem is where the number of cells is 250. However, as the number of cells and resolution decrease, the number of scenarios in which collision avoidance fails increases because it is difficult to accurately identify the surrounding situation. In addition, when the number of cells is 500, the surrounding situation is identified most accurately. However, the amount of computation increases exponentially. Therefore, learning is not completed normally.

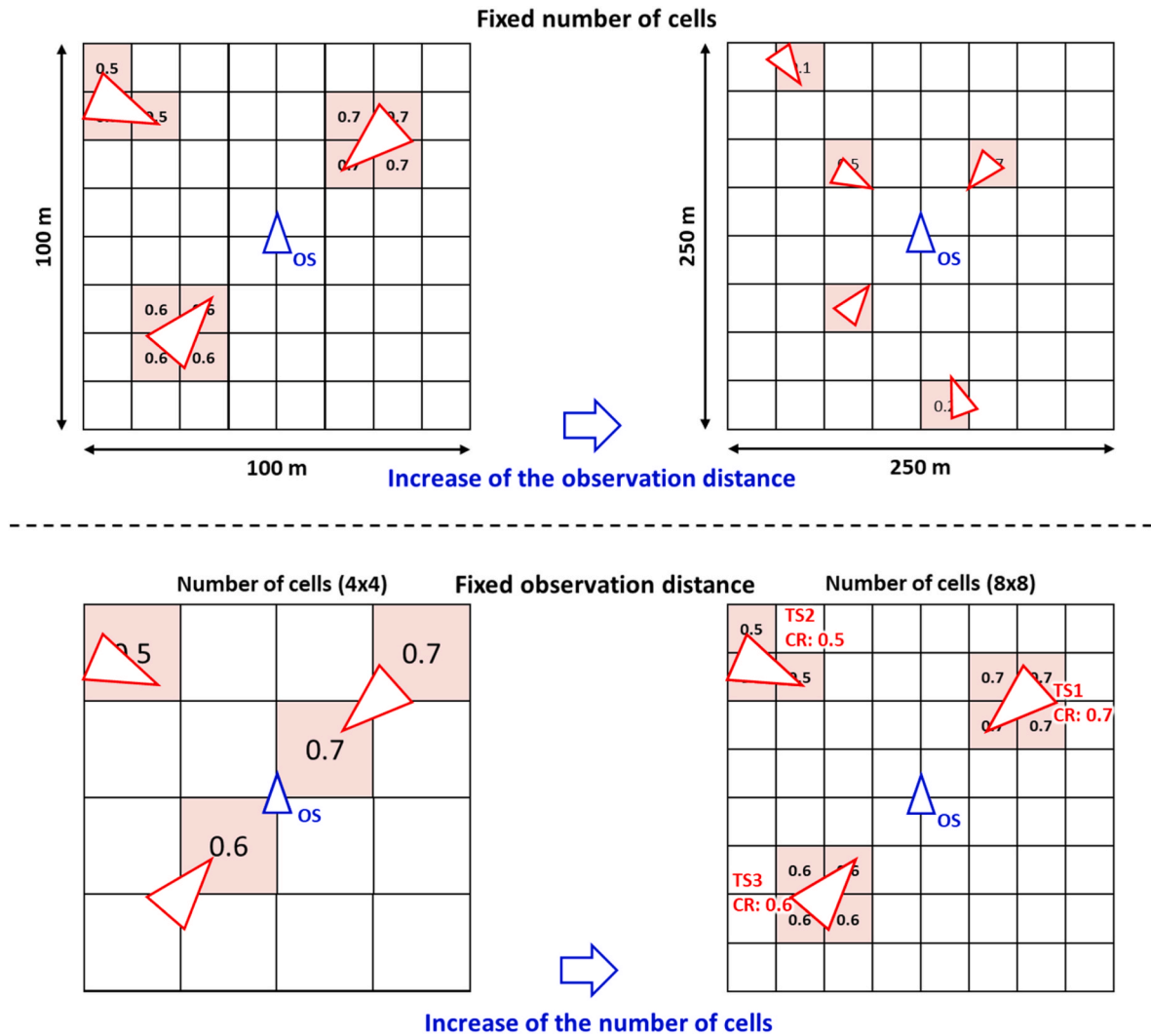


Fig. 10. Effect of the variation in each parameter.

Table 7

Parameter tuning for collision avoidance model.

Parameter	Value of the parameter	Fixed value	Cell size (l_c)
Observation distance (D)	100 m	Number of cells = 250	0.4 m
	250 m		1 m
	500 m		2 m
	1000 m		4 m
Number of cells (N)	50	Observation distance = 250 m	5 m
	100		2.5 m
	250		1 m
	500		0.5 m

4.3. Performance comparison with other methods

Next, a collision avoidance method that controls only the path of the OS (Chun et al., 2021) and the collision avoidance method that simultaneously controls the path and speed proposed in this study were applied to the Imazu problem and compared.

The existing method controls the rudder angle discretely and does not control the propeller RPM. Therefore, it is fixed at half of the maximum propeller RPM. Additionally, the method considers only the TS with the highest collision risk based on the MLP. In contrast, the new method proposed in this study simultaneously controls the path and speed by controlling the rudder angle and propeller RPM. Furthermore,

collision avoidance is performed by simultaneously inputting the position information of all the TSs within a certain distance based on the CNN state. The collision avoidance results for the Imazu problem are listed in Table 10.

In the existing method, collision avoidance failed in four scenarios of the Imazu problem (scenarios 6, 10, 13, and 20). The average maximum collision risk was 0.99, which was approximately 5% higher than that for the new collision avoidance method. The average minimum distance to the TS was 15.1 m, which was approximately 21% closer. In addition, this method did not control the speed of the OS by fixing the propeller RPM, and the speed error increased significantly compared with the new collision avoidance method. Similar to Section 4.1, we compared the path and speed for each method for scenarios 8, 10, 13, and 17 in the Imazu problem. Figs. 13 and 14 show a comparison of the collision avoidance results between path control and path-speed control.

In each scenario, because the previous method considered the TS with the highest collision risk, it failed to identify the TSs approaching from opposite directions simultaneously or avoid the collision (Imazu problem 10, 13). Even when it succeeded in avoiding the collision, it did not control the speed of the OS and could not reduce or increase the speed. Therefore, it was continuously exposed to hazardous situations and had a longer route (Imazu problem 8, 17). In addition, because we fixed the propeller RPM of the OS to half the maximum propeller RPM, we observed that the speed of the OS converged to approximately 0.6 m/s. This resulted in an error of approximately 40% from the target speed

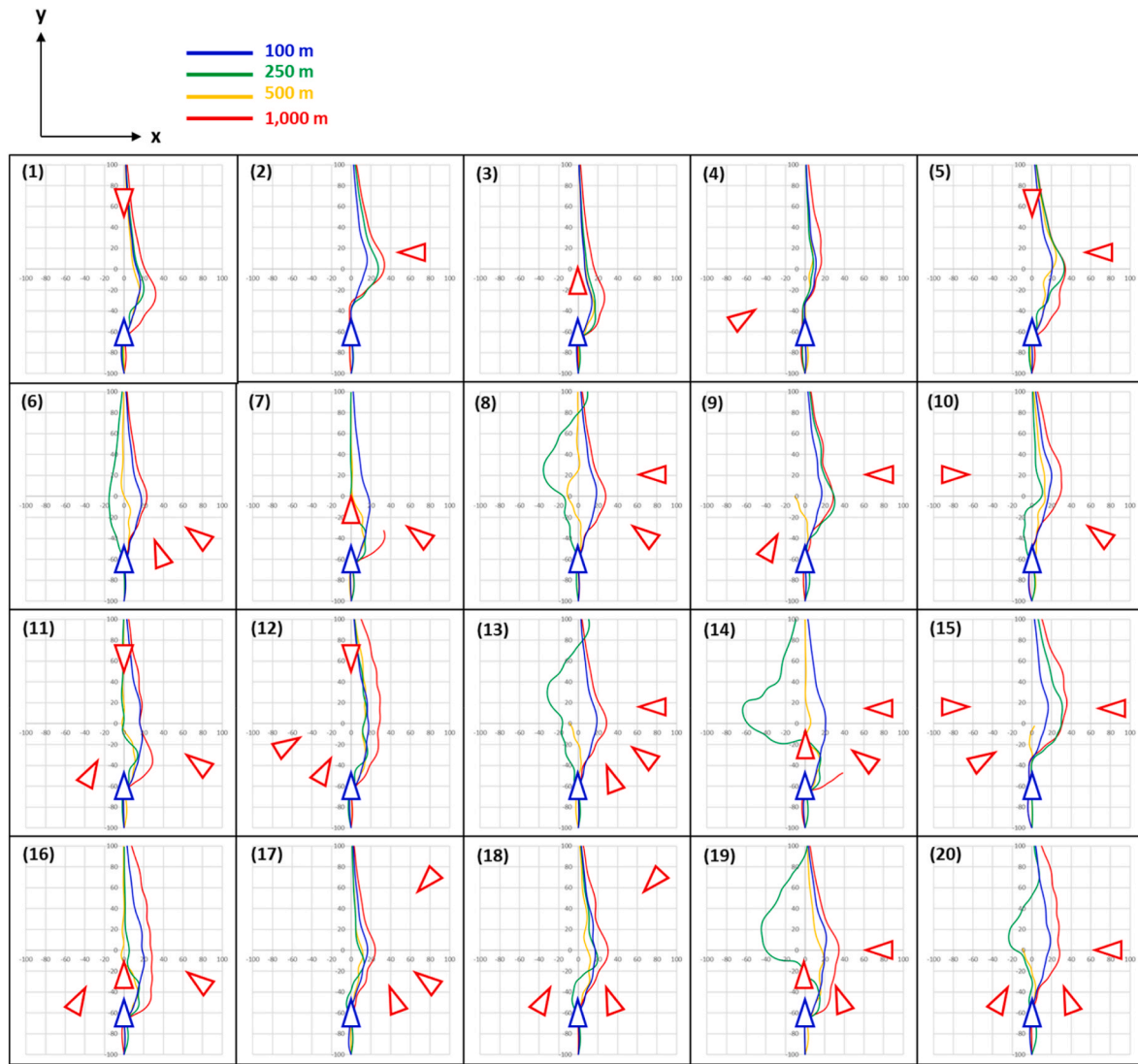


Fig. 11. Collision avoidance path for the scenarios according to the observation distance.

Table 8

Result of collision avoidance according to the observation distance.

Observation distance (D)	Number of successful scenarios	Average maximum collision risk of the TS	Average speed [m/s]	Average travel time [s]	Average travel distance [m]
100 m	20/20	0.88	0.87	236	206
250 m	20/20	0.87	0.80	275	220
500 m	18/20	0.91	0.82	253	212
1000 m	18/20	0.88	0.83	260	214

of 1 m/s. However, in this case, even without controlling the propeller RPM, the speed decreased marginally when the OS was turned. Meanwhile, in our proposed method, because we controlled the propeller RPM simultaneously, we performed sophisticated collision avoidance by reducing or increasing the speed to rapidly leave the hazardous situation. We observed that the speed of the OS approached the target speed after the collision avoidance situation ended.

Also, another collision avoidance method based on velocity obstacles (VO) was applied to the Imazu problem and compared. VO is a method that calculates the set of all velocities of a ship where that collision can occur and finds the optimal velocity to arrival which can avoid the

collision (Huang et al., 2019; C. Wang et al., 2023). The rudder angle and propeller RPM are controlled by the optimal velocity with the result of PID control. VO can provide an efficient route because the method finds the optimal velocity to arrival. However, the result of VO can be a potentially dangerous operation with a high overall collision risk. The collision avoidance results for the Imazu problem are listed in Table 11.

As a result of the VO-based collision avoidance method, collision avoidance succeeded in all scenarios of the Imazu problem. The average travel distance was 203 m, which was approximately 8% reduced compared with the DRL-based collision avoidance method. However, the average collision risk was 0.98, which was approximately 4% higher than that for the DRL-based collision avoidance method. Therefore, compared to the VO-based collision avoidance method, the DRL-based collision avoidance method is slightly more inefficient but results in a safer operation.

4.4. Complex test scenario for comparison between the two methods

In this study, to compare the collision avoidance performance of the two collision avoidance methods (path control and path-speed control) as described in Section 4.3, we applied these to the benchmark scenario of the Imazu problem and compared the results. However, the Imazu problem is a simple example of collision avoidance for a maximum of

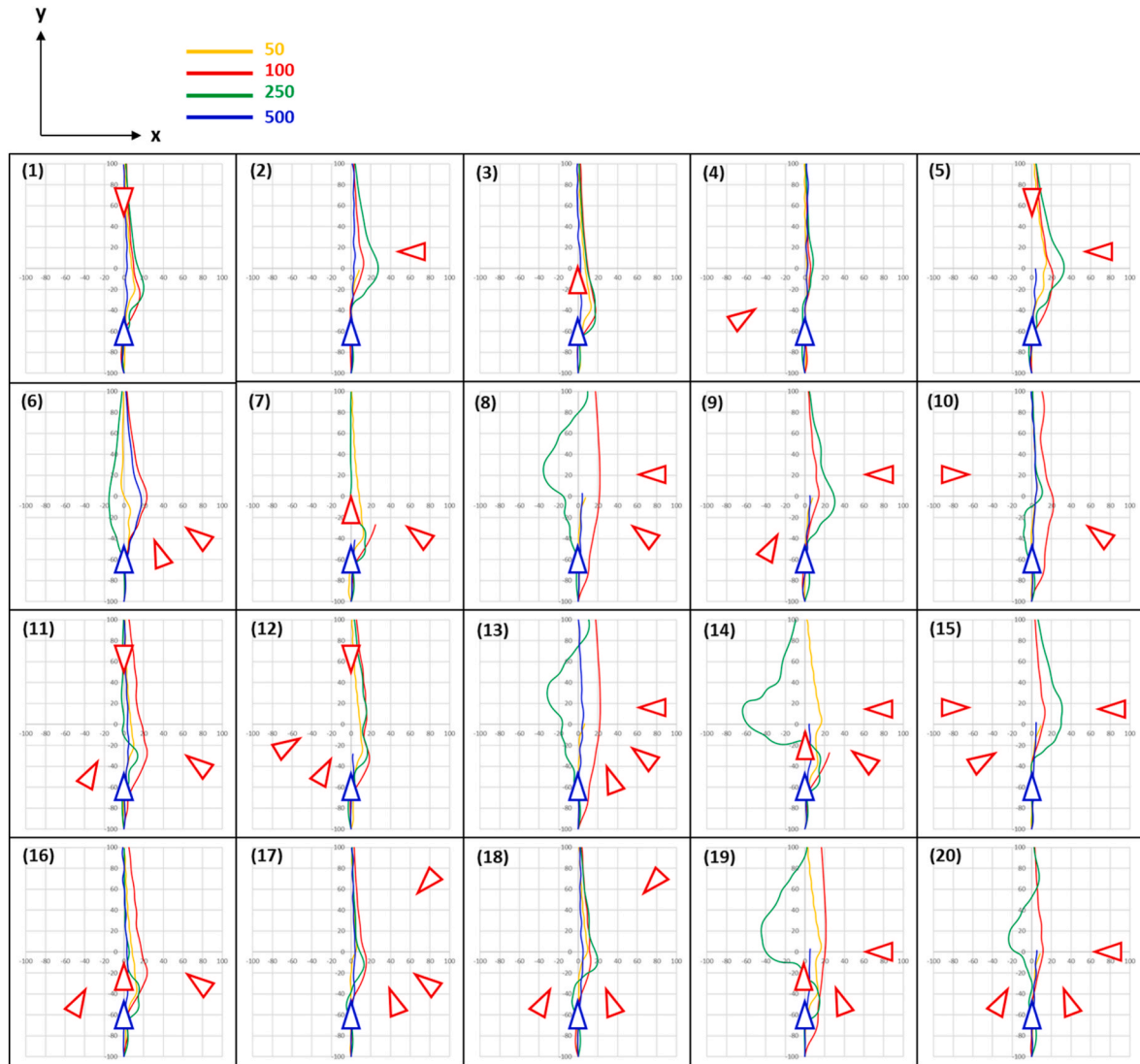


Fig. 12. Collision avoidance path for the scenarios according to the number of cells.

Table 9

Result of collision avoidance according to the number of cells.

Number of cells (N)	Number of successful scenarios	Average maximum collision risk of the TS	Average speed [m/s]	Average travel time [s]	Average travel distance [m]
50	16/20	0.94	0.81	250	203
100	18/20	0.88	0.68	317	205
250	20/20	0.87	0.80	275	220
500	11/20	0.96	0.91	222	201

Table 10

Collision avoidance result between path control and path-speed control.

	Number of success scenario	Average of maximum collision risk of the TS	Average of minimum distance of the TS [m]	Average speed [m/s]
Path control	16/20	0.99	22.1	0.49
Path-speed control	20/20	0.94	25.2	0.79

three TSs. It does not represent all the situations that may occur in real-world operations. Therefore, in this study, we compared the collision avoidance results of path control and path-speed control for a complex test scenario in which multiple TSs operate around an OS, thereby simulating real-world operations. Fig. 15 shows a complex test scenario for a comparison between the two methods.

In addition, when only path control was performed, the cases with propeller RPMs of 300 (which is half the maximum) and 600 (which is the maximum) were compared. Table 12 lists the collision avoidance results for the complex test scenarios.

When only path control was performed, we observed that collisions occurred (TS 3 for propeller RPM of 300 and TS 5 for propeller RPM of 600) when multiple TSs were approaching the OS. This occurred because the OS could not perform sophisticated collision avoidance. However, when path-speed control was performed, we observed that collision avoidance was performed successfully for many TSs and the OS reached the destination safely. Figs. 16 and 17 show the collision avoidance results for the complex test scenario.

Next, we compared the collision avoidance paths of the methods for two scenarios with 10 randomly placed TSs. Figs. 18 and 19 show two random test scenarios for a comparison between the two methods.

Table 13 lists the collision avoidance results for the random test #1 scenario.

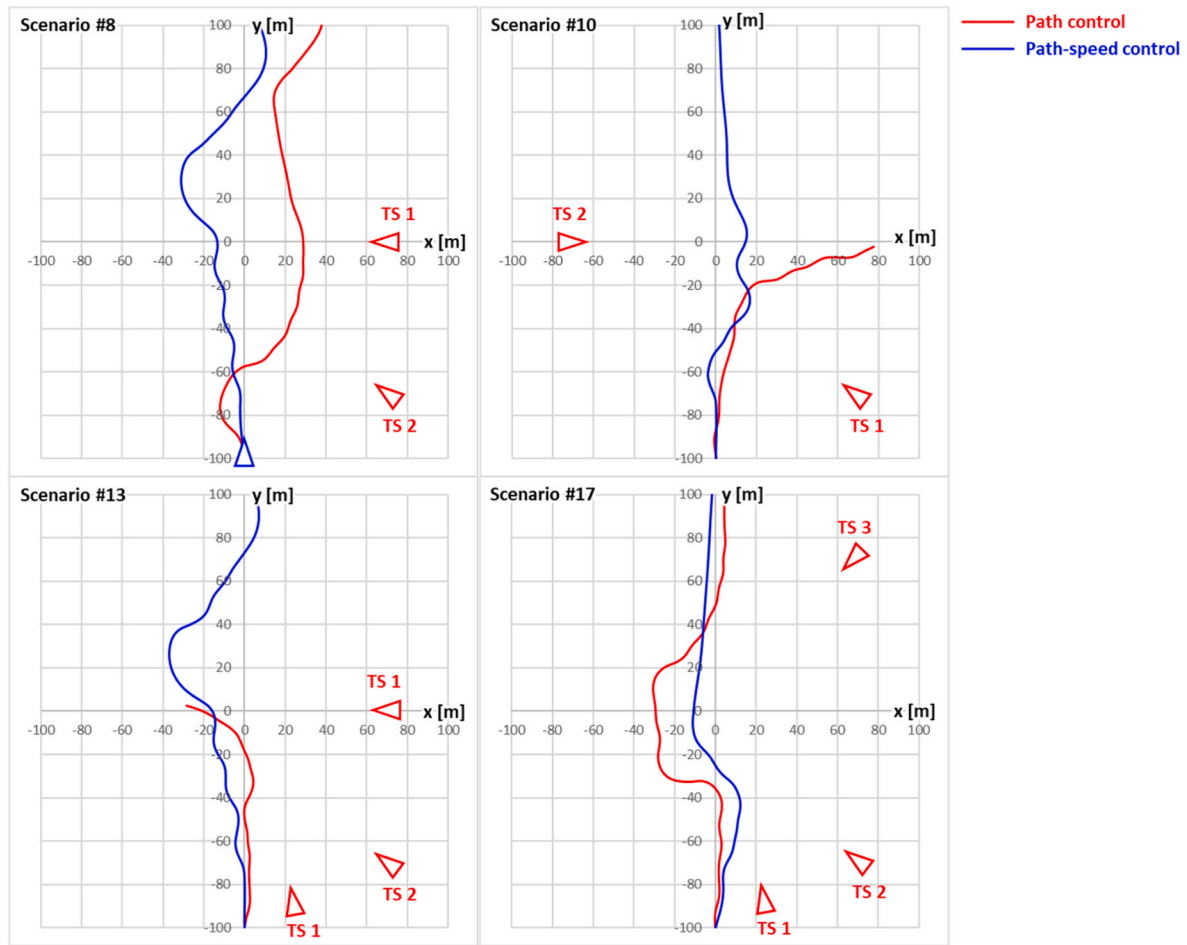


Fig. 13. Comparison of collision avoidance path between path control and path-speed control.

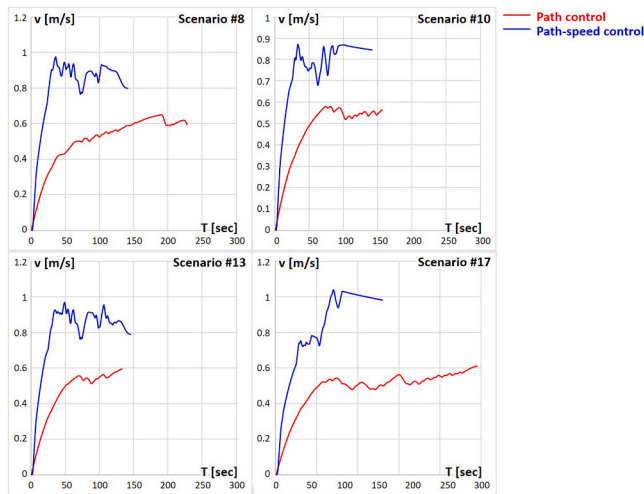


Fig. 14. Comparison of the speed of the OS between path control and path-speed control.

In this scenario, all three controls successfully avoided the TSs. In this case, the maximum collision risk of the TS was measured to be about 0.99 with path control, and it was slightly reduced to 0.98 with path-speed control. In addition, the path control did not reach the destination to avoid TS 8 because it did not control the speed of the OS in the case of propeller RPM of 600. In the same situation, the path-speed

Table 11

Collision avoidance result between the DRL and VO.

	Number of success scenario	Average of maximum collision risk of the TS	Average of minimum distance of the TS [m]	Average travel distance [m]
VO	20/20	0.98	17.0	203
DRL	20/20	0.94	25.2	220

control reduced the speed of the OS and reached the destination after TS 8 passed over the OS. Figs. 20 and 21 show the collision avoidance results for the random test #1 scenario.

Table 14 lists the collision avoidance results for the random test #1 scenario.

In this scenario, all three controls successfully avoided the TSs. In all three controls, the maximum collision risk of the TS was measured to be 0.99, but the path-speed control had the largest minimum distance. Also, similar to the previous scenario, the path control was successful in avoiding collisions with the TS but ended the scenario far from the destination because it does not control the speed of the OS. On the other hand, path-speed control succeeded in reaching the destination. Figs. 22 and 23 show the collision avoidance results for the random test #2 scenario.

Finally, the control results were compared to the case where there was a continuous randomized change in the movement of the TSs. In this case, the initial positions, velocities, and orientations of the TSs are shown in Fig. 15.

Table 15 lists the collision avoidance results for the random test #1 scenario. At a propeller RPM of 300, the OS can't avoid collision with the

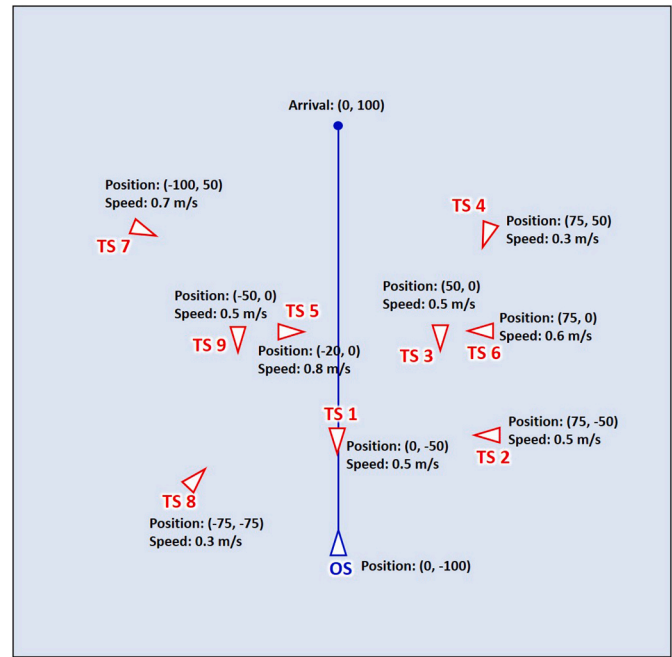


Fig. 15. Complex test scenario for comparison between two methods.

Table 12
Collision avoidance result for complex test scenario.

	Maximum collision risk of the TS	Minimum distance of the TS [m]	Average speed [m/s]
Path control (Propeller RPM of 300)	1 (Collision)	11.4 m	0.40
Path control (Propeller RPM of 600)	1 (Collision)	7.6 m	0.94
Path-speed control	0.98	19.5 m	0.88

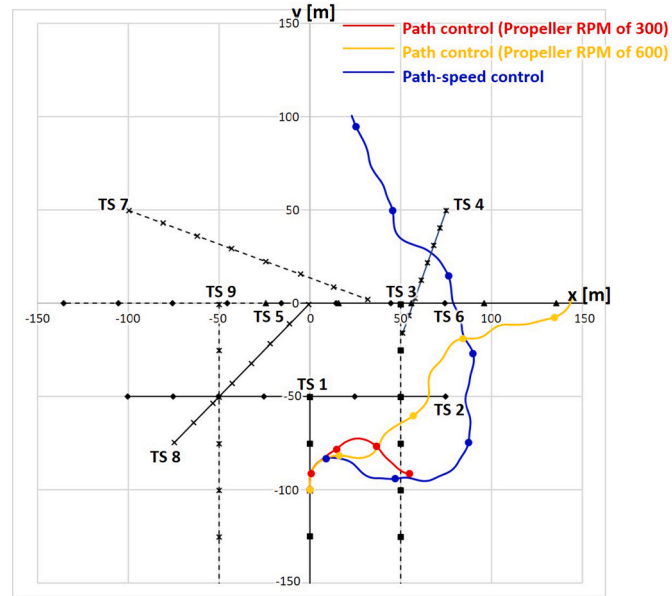


Fig. 16. Comparison of collision avoidance path for complex test scenario.

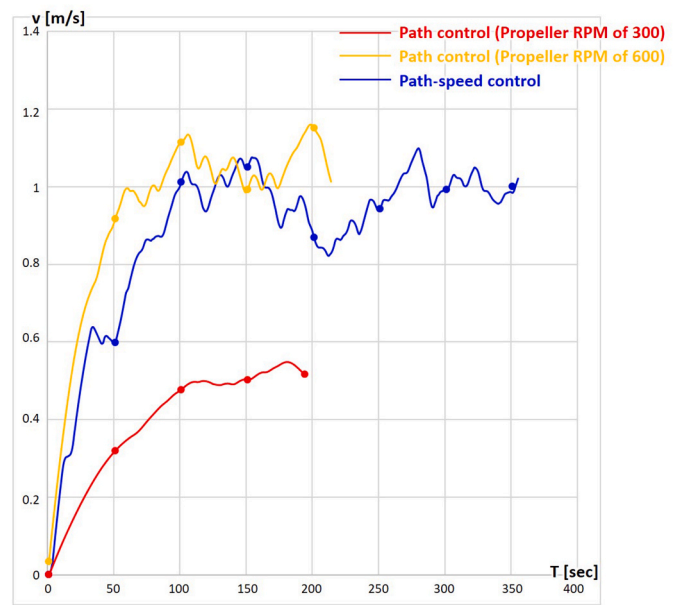


Fig. 17. Comparison of the speed of the OS for complex test scenario.

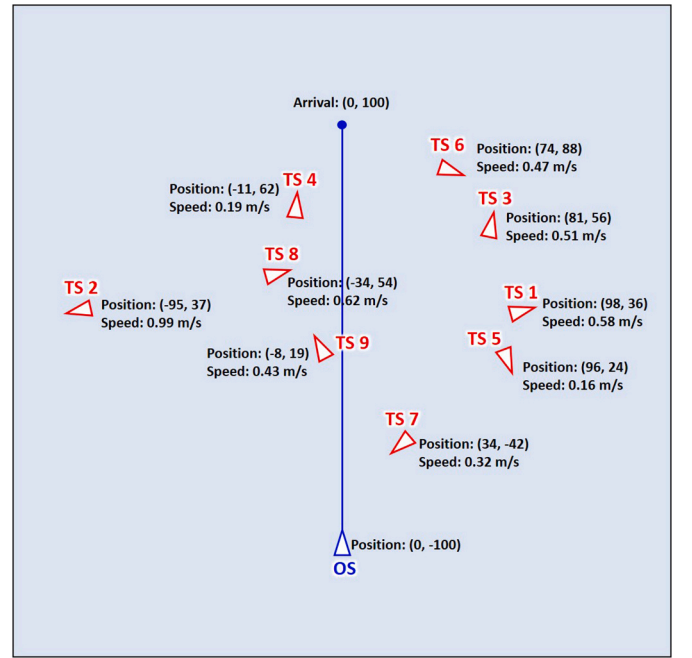


Fig. 18. Random test #1 scenario for comparison between two methods.

TS, and at a propeller RPM of 600, the OS spun around in an unnatural motion. On the other hand, the path-speed control reached the destination with an efficient path. Figs. 24 and 25 show the collision avoidance results with the random movement of TSs.

5. Conclusion and future work

In this study, we defined MLP-CNN states to simultaneously identify multiple TSs around the OS and perform collision avoidance. Additionally, we proposed a DRL-based collision avoidance method that defines the rudder angle and propeller RPM of the OS as actions. We also defined an efficiency reward for efficient operation of the OS and a safety reward for safe operation between the OS and TSs. In this process, the quantitative collision risk between the OS and TS was calculated

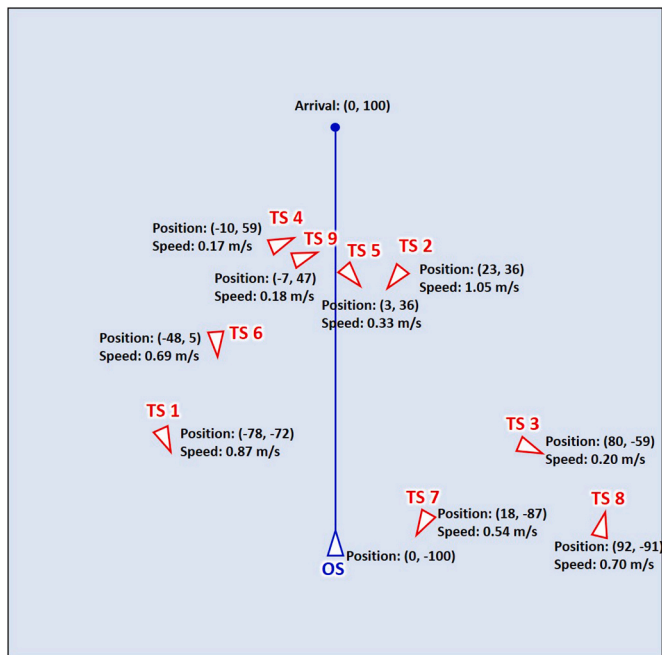


Fig. 19. Random test #2 scenario for comparison between two methods.

Table 13
Collision avoidance result for the random test #1 scenario.

	Maximum collision risk of the TS	Minimum distance of the TS [m]	Average speed [m/s]
Path control (Propeller RPM of 300)	0.99	17.9 m	0.52
Path control (Propeller RPM of 600)	0.99	16.4 m	0.99
Path-speed control	0.98	22.0 m	0.87

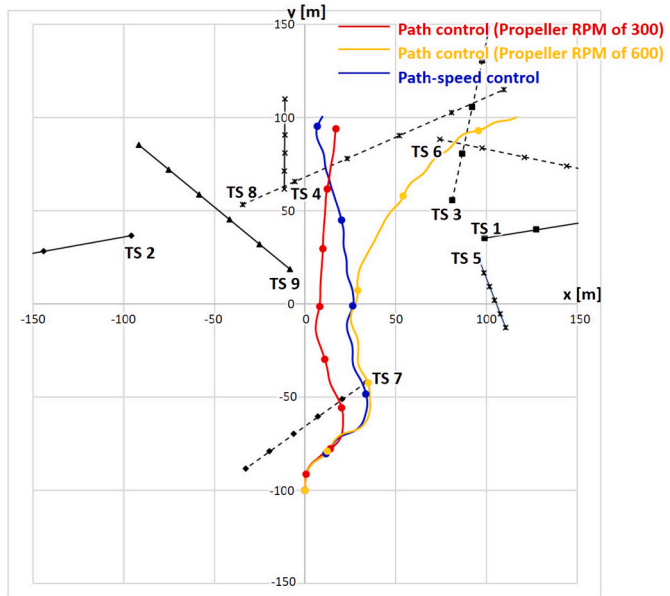


Fig. 20. Comparison of collision avoidance path for random test #1 scenario.

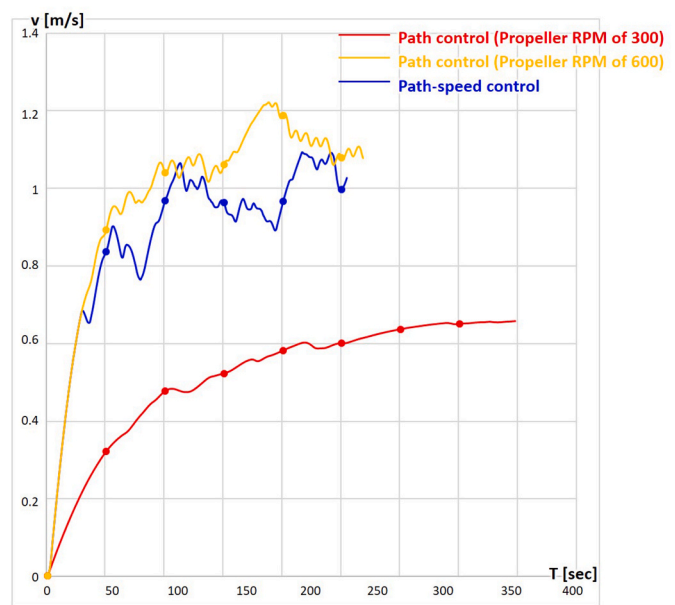


Fig. 21. Comparison of the speed of the OS for random test #1 scenario.

Table 14
Collision avoidance result for the random test #2 scenario.

	Maximum collision risk of the TS	Minimum distance of the TS [m]	Average speed [m/s]
Path control (Propeller RPM of 300)	0.99	4.4 m	0.51
Path control (Propeller RPM of 600)	0.99	8.4 m	0.98
Path-speed control	0.99	11.0 m	0.74

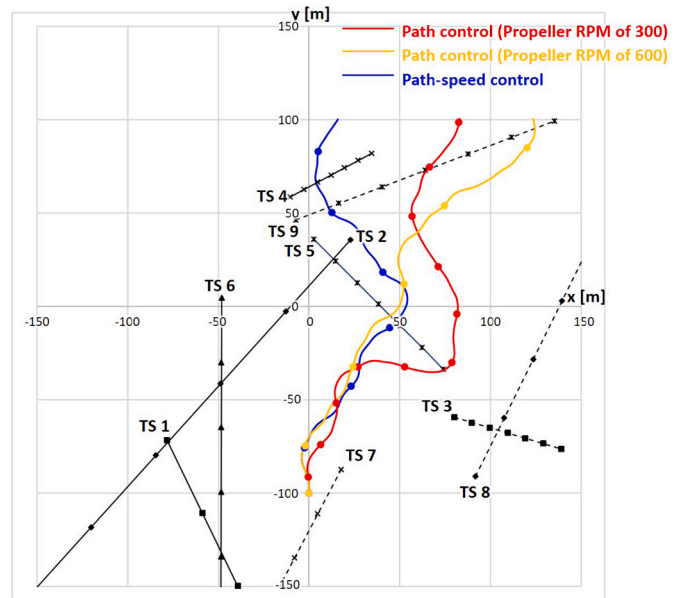


Fig. 22. Comparison of collision avoidance path for random test #2 scenario.

using the ship domain and CPA, and the rewards were defined to comply with the COLREGs rules.

We applied the proposed method to the Imazu problem with 20

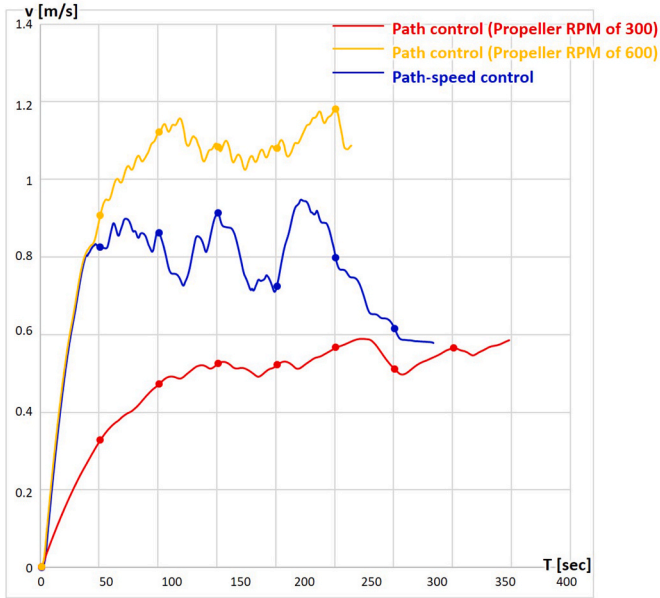


Fig. 23. Comparison of the speed of the OS for random test #2 scenario.

Table 15

Collision avoidance results with the random movement of TSs.

	Maximum collision risk of the TS	Minimum distance of the TS [m]	Average speed [m/s]
Path control (Propeller RPM of 300)	1 (Collision)	7.4 m	0.40
Path control (Propeller RPM of 600)	0.99	10.2 m	0.94
Path-speed control	0.98	12.6 m	0.88

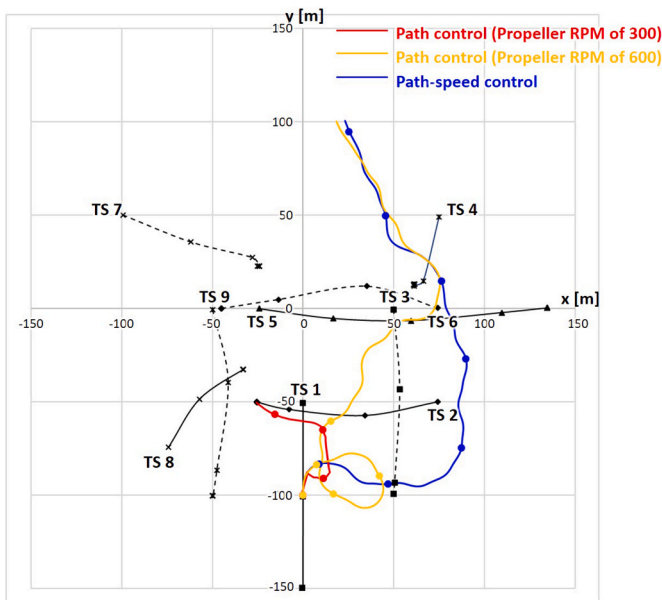


Fig. 24. Comparison of collision avoidance path with the random movement of TSs.

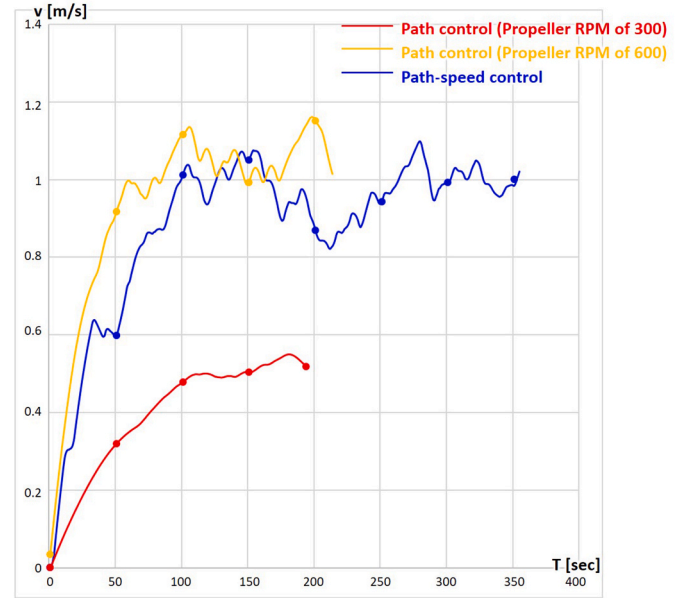


Fig. 25. Comparison of the speed of the OS with the random movement of TSs.

benchmark scenarios to evaluate the avoidance performance based on the learning level. We observed that collision avoidance was successful in 2, 4, 11, and 20 scenarios as the learning progressed to 10, 100, 1,000, and 10,000 iterations for a total of 20 scenarios. Moreover, collision avoidance was successful in all the scenarios. We also compared the collision avoidance performance based on the parameter tuning of the CNN state defined in Section 3.1. We trained the collision avoidance model by varying two parameters: the observation distance and the number of cells. We observed that collision avoidance was successful for all the scenarios when the observation distance was 100 m and 250 m, and number of cells was 250. We also observed that the average maximum risk was lowest when the observation distance was 250 m and number of cells was 250. This was approximately 4.4% less than the highest value of 0.91 (500 m) for the observation-distance result and approximately 15.6% less than the highest value of 0.96 (500) for the number-of-cells result.

In addition, we verified the difference in the collision avoidance paths when path speed control and simple path control were performed. For the 20 scenarios of the Imazu problem, when only path control was performed, collisions occurred in four scenarios, and safe operation failed. The average maximum collision risk was 0.99. However, when path-speed control was performed, all the scenarios avoided collisions successfully and performed safe operations, with an average maximum collision risk of 0.94 (a reduction of approximately 5%). Similarly, we compared the proposed method with the VO-based collision avoidance method. For 20 scenarios of the Imazu problem, the average collision risk and the average travel distance were 0.98 and 203 m. However, when the proposed method was performed, the average collision risk and the average travel distance were 0.94 (a reduction of approximately 4%) and 220 m (an increase of approximately 8%). Therefore, compared to the VO-based collision avoidance method, the DRL-based collision avoidance method is slightly more inefficient but results in a safer operation. Also, for the complex test scenarios with multiple TSs, performing only path control resulted in collisions or inefficient control that prevented the OS from reaching its destination. However, when path-speed control was performed, safe operations that avoided collisions were performed successfully. With path-speed control, the OS will maintain the desired target speed if there is no TS nearby. If there is a TS with a higher collision risk than $CR_{allowable}$ in the vicinity of the OS, the OS will avoid that TS based on the COLREGs.

The COLREGs basically assume to avoid a one-on-one encounter

situation. However, in many of the scenarios proposed in this study, a situation is created where multiple TSs operate near the OS at the same time. At this time, depending on the sum of various rewards, including the COLREGs rewards, if the collision risk is very high and a very long distance must be traveled according to the COLREGs, a more efficient and safer path can be selected even if it violates the COLREGs slightly. Therefore, if there are multiple TSs in the vicinity, all of them are considered simultaneously, and a safer path may be chosen, even if it violates the COLREGs. In addition, since the method proposed in this study simultaneously controls the path and speed, the speed can be flexibly controlled to operate a safe path.

In future research, we would study methods to calculate the collision risk by considering the information uncertainty. Various errors are likely to occur in the TS information during the actual operation of an OS. Also, the collision risk assessment proposed in this study has the disadvantage that the collision risk changes very significantly when the relative speed between the two ships changes continuously and the TCPA changes continuously near 0. Therefore, as a future work, we are planning a study to more accurately quantify the threat between ships based on the modification of the fundamental calculation of collision risk rather than adjusting the collision risk through f_{angle} . Moreover, the abrupt variations in the collision risk caused by these can be considered to more accurately calculate collision risk and generate collision avoidance paths.

In this study, we introduced the efficiency reward to improve path following performance and the safety reward to improve collision avoidance performance. The rewards are divided into a total of 5 rewards. Then, we tried to maximize the total sum of the rewards. In the future, we plan to study how to give weights to each reward optimally based on the operation purpose. For example, if we need a more optimal operation from an economic perspective, we can give weight to the efficiency reward to better follow the route. On the other hand, if we need a safer operation, we can give weight to the safety reward to perform an operation that travels further but has a lower collision risk of the TS.

Also, we applied the proposed method to a small ship with high maneuvering performance, and the proposed method did not consider the reward for course change. This can cause the OS to turn frequently, which may have a significant impact on the operational performance of the OS. Therefore, the collision avoidance method proposed in this study will be trained on large ships such as LNG (Liquefied Natural Gas) carriers and VLCCs (Very Large Crude oil Carriers), and the results will be validated. Also, the reward to limit excessive course change of the OS will be considered. In this process, we will perform parameter tuning on the observation distance, number of cells, and cell size described in Section 4.2, and learn the optimal collision avoidance model by finding the optimal parameters for large ships.

Finally, the DRL used in this study is most commonly applied to autonomous driving and navigation in cars, USVs, and UAVs. However, there are several other learning-based autonomous driving approaches besides the DRL, such as learning controllers with MPC (Lefevre et al., 2016) or data-based learning (Caesar et al., 2019), and a comparative analysis with these methods will also be performed in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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